



Scientific research project

Mapping the Heat Divide: Spatio-Temporal Analysis of Thermal Hotspots and Socio-Economic Inequality in Dhaka (2014–2024)

Submitted to
Prof. Dr. Jan-Peter Mund
Chair of GIS and Remote Sensing
HNEE, Eberswalde

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Submitted by
Mahbubul Alam
Masters in Forest Information Technology
Matrikel Nr. 24215867

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Contents

List of Figures	5
Abbreviations.....	7
Abstract.....	8
Executive Summary.....	9
Acknowledgements.....	10
AI-Assistance Statement	11
Introduction	12
1.1 Background	12
1.2 Urban heat island (UHI) and thermal inequality in Dhaka	12
1.3 Problem statement	13
1.4 Research Questions	13
1.5 Objectives of the Study	13
1.6 Significance of the Study.....	14
1.7 Report Structure	14
2. Literature Review	15
2.1 Urban Heat Islands and LST Dynamics in Dhaka	15
2.2 Land-Cover Change and Its Influence on Surface Temperature	16
2.3 Heat Vulnerability, Poverty, and Public Health.....	16
2.4 LST Retrieval Methods and Time-Series Approaches	17
2.5 Mitigation Strategies and Policy Context.....	17
2.6 Synthesis and Justification for the Present Study.....	17
2.7 Research Gap	18
3. Conceptual Framework.....	19
4.Study Area.....	21
4.1 Geographic Setting of Dhaka	21
4.2 Climate and Seasonal Characteristics	21
4.3 Urbanization Trends and Land Cover Change.....	22
4.4 Socio-Economic Profile and Poverty Distribution	23
4.5 Rationale for Selecting Dhaka	23
5. Data and Materials.....	24
5.1 Landsat 8/9 Thermal Data (2014–2024)	24
5.2 MODIS LST Data (MOD11A2, 2014–2024)	25
5.3 Socioeconomic Data and Proxy Variables.....	25
5.4 Administrative Boundaries and Ancillary Data	26
5.5 Software and Tools	26

5.6 Spectral Indices (NDVI, NDBI, MNDWI).....	27
5.7 Data Pre-Processing Summary.....	27
6.Methodology.....	28
6.1 Overview of the Workflow.....	28
6.2 LST Retrieval.....	29
6.3Annual and Seasonal LST Composite Generation	29
6.4 Trend Analysis	30
6.5 Hotspot and Cold-Spot Detection	30
6.6 Poverty–Heat Linkage	31
6.7 Statistical Analysis Framework.....	31
6.8 Ethical Considerations.....	32
6.9 Methodological Limitations	32
7.Result	32
7.1 Temporal Patterns of Land Surface Temperature (2014–2024)	32
7.2 Spatial and Seasonal Patterns of Land Surface Temperature (2017)	34
7.3 Trend Analysis of Land Surface Temperature (2014–2024)	37
7.4 Long-Term LST Trends and Urban Heat Expansion.....	40
7.5 Spatial Distribution of Hotspots and Cold Spots.....	45
7.6 Spatial Distribution of Poverty (2017)	48
7.7 Heat–Poverty Interaction.....	50
Identification of High-Risk Zones	56
Supplementary Statistical Evidence of Thermal Inequality	57
8. Discussion.....	58
9. Limitations and Future Work	59
9.1 Limitations.....	59
9.2 Future Work	59
10. Conclusion.....	60
11. References	61
Appendix	64

List of Figures

No	Caption	Page
Figure 1.1	Study Workflow Overview	11
Figure 2.1	Systematic Review Flowchart	12
Figure 3.1	Conceptual Framework for Heat–Poverty Interaction	16
Figure 4.1	Map of Dhaka Metropolitan Area	17
Figure 4.2	Land Cover Mapping of Dhaka, Bangladesh	18
Figure 4.3	Dhaka poverty Map 2017	19
Figure 5.1	Presents the temporal distribution of Landsat observations	20
Figure 6.1	Processing Workflow Diagram	24
Figure 6.1c	Annual LST Distribution for Dhaka (2014–2024)	29
Figure 6.2	sensor wise images and their cloud percentage	25
Figure 6.3	Trend Analysis Workflow MK + Theil–	26
Figure 6.4	Heat–Poverty Linkage Workflow (own presentation)	27
Figure 6.6a	also serves as the administrative reference map for Dhaka, providing ward labels that support	45
Figure 6.6b	indicates that several wards exhibit markedly higher	45
Figure 7.1b	Monthly mean LST heatmap (2014–2024). Clear seasonal cycles appear, with cooler	28
Figure 7.1d	Decadal LST Variation (2014–2024)	30
Figure 7.2a	presents zone-wise monthly LST maps for January–May and October–December, using a	31
Figure 7.2b	illustrates seasonal LST variation across Dhaka, highlighting only zones that exceed the	31
Figure 7.2c	shows monthly minimum, mean, and maximum LST values for 2017. Temperatures peak in	32
Figure 7.2d	Zone-Wise Δ LST (April–December)	32
Figure 7.3a	Mann–Kendall Trend Significance (2014–2024)	34
Figure 7.3b	Theil–Sen Slope of LST Change ($^{\circ}$ C/year), 2014–2024	35
Figure 7.4	b) highlight the seasonal peak of thermal stress. Over	37
Figure 7.4b	Pre-Monsoon LST Maps (2014–2024)	37
Figure 7.4c	LST, Satellite Imagery & Heat-Risk Comparison (2014 vs 2020)	38
Figure 7.4d	Heat–Poverty Vulnerability Map of Dhaka (left) and LST Distribution for April 2017 (right)	39

Figure 7.4e	Pre_Monsoon NDVI Maps (2014–2024)	39
Figure 7.4f	NDVI–LST Scatter Plot (2014–2024)	40
Figure 7.4g	NDVI & LST Dual-Axis Trend Plot (2014–2024)	40
Figure 7.5c	Structural patterns of hotspot persistence and cooling deficits in Dhaka (2014–2024)	43
Figure 7.6a	Spatial Distribution of Poverty 2017	44
Figure 7.6b	HPVI (Building Density- NB_BUDENS)	45
Figure 7.6c	Spatial Distribution and Persistence of Socioeconomic Vulnerability in Dhaka	46
Figure 7.7	f: Comparison of Poor vs Non-Poor Areas	50
Figure 7.7a	presents the correlation matrix heatmap, summarizing the pairwise relationships among the	46
Figure 7.7b	displays the scatter plot between Normalized Difference Vegetation Index (NDVI) and	47
Figure 7.7c	presents the scatter plot between NDBI and LST ($LST \geq 20\text{ }^{\circ}\text{C}$). The regression slope is	48
Figure 7.7d	illustrates the scatter plot between MNDWI and LST ($LST \geq 20\text{ }^{\circ}\text{C}$). The regression slope	48
Figure 7.7e	presents the scatter plot between the poverty index (proxied by building density) and LST	49
Figure 7.7f	presents box plots comparing poor and non-poor wards across four key indicators. Poorer	50
Figure 7.7g	Multi-panel poverty correlations plots	51
Figure 7.7h	Poverty vs Heat correlation heatmap	51
Figure 7.7i	Distribution of Poverty Index and Average LST illustrates the underlying data	52
Figure 7.7j	HPVI quantile map	53

Abbreviations

Abbreviation	Full Form
LST	Land Surface Temperature
UHI	Urban Heat Island
SUHII	Surface Urban Heat Island Intensity
LULC	Land Use / Land Cover
NDVI	Normalized Difference Vegetation Index
NDBI	Normalized Difference Built-up Index
MNDWI	Modified Normalized Difference Water Index
IBI	Index-Based Built-up Index
HPVI	Heat-Poverty Vulnerability Index
HVI	Heat Vulnerability Index
MK	Mann-Kendall Test
TS	Theil-Sen Slope Estimator
TOA	Top of Atmosphere
BT	Brightness Temperature
SC Algorithm	Single Channel Algorithm
DEM	Digital Elevation Model
GEE	Google Earth Engine
MODIS	Moderate Resolution Imaging Spectroradiometer
MOD11A2	MODIS Land Surface Temperature 8-Day Composite Product
EO4SD	Earth Observation for Sustainable Development
NB_BUDENS	Normalized Building Density (EO4SD Urban Morphology)
HIES	Household Income and Expenditure Survey
OSM	OpenStreetMap

Abstract

Rapid urbanisation in Dhaka has intensified land surface temperature (LST) patterns, increased heat exposure and disproportionately affecting socio-economically vulnerable communities. This study analyses the spatio-temporal dynamics of LST from 2014 to 2024 using Landsat 8/9 thermal imagery, complemented by long-term MODIS data. Seasonal composites, Mann-Kendall and Theil-Sen trend analysis, and hotspot detection were applied to characterise thermal behaviour across the city. Building density from the EO4SD urban morphology dataset was used as a proxy for poverty to assess heat-vulnerability interactions at the ward level.

Results reveal a statistically significant warming trend and a clear expansion of thermal hotspots, particularly in dense residential and industrial zones. Vegetated and water-rich areas consistently exhibit lower temperatures, while built-up intensity strongly correlates with elevated LST. High-density wards experience greater heat exposure, indicating a strong heat-poverty relationship.

The study identifies several high-risk zones where long-term warming, hotspot concentration, and socio-economic vulnerability converge. These findings highlight the need for targeted heat-mitigation strategies and climate-resilient urban planning. The research offers a reproducible framework for assessing urban heat in data-scarce megacities and contributes to climate-equity discussions in rapidly urbanising regions.

Executive Summary

Dhaka's rapid urban expansion has intensified heat exposure across the city. This study examines how land surface temperature (LST) has changed between 2014 and 2024, how seasonal and spatial patterns vary, and how heat exposure aligns with socio-economic vulnerability at the ward level.

Objectives

1. Analyse spatio-temporal LST patterns (2014–2024).
2. Assess long-term warming trends using MODIS.
3. Identify thermal hotspots and cold spots.
4. Use building density as a proxy for poverty.
5. Identify high-risk zones for targeted interventions.

Methods

The analysis was conducted using the Google Earth Engine Python API. Multi-year Landsat and MODIS observations were processed to generate annual and seasonal LST composites and evaluate long-term warming trends. Dhaka's GADM ward boundaries were used as the administrative framework, and the NB_BUDENS building-density layer from the EO4SD urban morphology dataset was clipped to the study area. Zonal statistics were applied to derive ward-level building-density values, enabling the use of building density as a proxy for socio-economic vulnerability and supporting the assessment of heat–poverty interactions.

Key Findings

1. Dhaka shows a clear warming trend, strongest in dense residential and industrial areas.
2. Pre-monsoon months exhibit the highest heat-stress levels.
3. Thermal hotspots expanded significantly from 2014 to 2024.
4. Vegetation and water bodies consistently reduce surface temperatures.
5. High-density wards are systematically hotter, confirming a heat–poverty divide.
6. High-risk zones include Mirpur, Tejgaon, Mohammadpur, Rampura, and Jatrabari.

Implications

Targeted greening, heat-resilient infrastructure, and prioritising vulnerable wards are essential to reduce rising heat risks. Integrating heat-risk mapping into urban planning can support more equitable and climate-resilient development across Dhaka.

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I also acknowledge the open-access platforms and institutions whose datasets made this research possible, including USGS, NASA, Google Earth Engine, and the EO4SD Urban Morphology project. Their commitment to accessible geospatial data continues to empower researchers worldwide.

AI-Assistance Statement

This report was prepared with the assistance of digital writing tools. Grammarly was used to refine grammar and readability, while Microsoft Copilot supported the structuring of sections and the organisation of ideas. All analysis, interpretations, coding, figures, and scientific conclusions are entirely the author's own work.

Introduction

1.1 Background

Land Surface Temperature (LST) is a key indicator of surface energy balance and environmental change, reflecting how different land-cover types absorb, store, and release heat (Li et al., 2013). In rapidly urbanising regions, LST is strongly influenced by the replacement of natural surfaces with impervious materials such as concrete, asphalt, and metal roofing, which store heat efficiently and elevate daytime and nighttime temperatures (Oke et al., 1982; Santamouris, 2014).

Dhaka exemplifies this transformation. Over recent decades, the city has expanded rapidly and often without coordinated planning, resulting in extensive loss of vegetation and wetlands (Dewan et al., 2009; Imran et al., 2021). Multiple studies have documented rising surface temperatures and intensifying Urban Heat Island (UHI) effects across the city (Dewan et al., 2021; Faisal et al., 2022). These changes are particularly concerning given Dhaka's high population density and the large share of residents living in low-income or informal settlements with limited adaptive capacity (Hossain et al., 2022).

Recent advances in cloud-based geospatial processing, including semi-automatic LST workflows, now enable consistent multi-year thermal analysis (Mohiuddin & Mund, 2024). Building on these developments, this study examines Dhaka's thermal dynamics over a decade and links them with socio-economic vulnerability to better understand spatial inequality in heat exposure.

1.2 Urban heat island (UHI) and thermal inequality in Dhaka

The UHI effect describes the tendency of urban areas to remain warmer than their surroundings due to reduced vegetation, increased impervious surfaces, and anthropogenic heat emissions (Arnfield, 2003). In Dhaka, UHI intensity has increased steadily over the past two decades, with satellite and meteorological observations showing clear seasonal variability and stronger contrasts during cooler months (Dewan et al., 2021; Jiang et al., 2024).

However, the impacts of urban heat are not evenly distributed. Wealthier neighbourhoods typically benefit from better building materials, more vegetation, and improved ventilation, while low-income and informal settlements—home to nearly 40% of Dhaka's population—are characterised by dense structures, limited airflow, and minimal tree cover (Hossain et al., 2022; Dewan et al., 2009). These environments trap heat and offer limited relief during hot days and warm nights. Evidence from Bangladesh and comparable contexts shows that heat stress disproportionately affects outdoor workers, informal labourers, and residents of substandard housing (Nahar et al., 2010; Kjellstrom et al., 2009).

This uneven exposure creates thermal inequality, where socio-economically disadvantaged groups face greater heat-related risks. Despite growing recognition of this issue, Dhaka lacks city-wide, spatially explicit evidence linking thermal patterns with poverty. Addressing this gap is essential for climate-equitable planning and aligns with national and municipal adaptation priorities, including Bangladesh's National Adaptation Plan (Government of Bangladesh, 2023) and Dhaka's emerging Climate Action Plans.

1.3 Problem statement

Although Dhaka's Urban Heat Island (UHI) and land surface temperature (LST) dynamics have been widely studied, existing research has focused primarily on physical drivers such as land-cover change, vegetation loss, and impervious surface expansion (Imran et al., 2021; Faisal et al., 2022; Kafy et al., 2021). Far less attention has been given to the social dimension of heat exposure, particularly how thermal patterns intersect with socio-economic vulnerability.

Current studies also rely heavily on limited satellite dates or short time periods, restricting insight into long-term thermal behaviour (Trotter et al., 2017; Sresto et al., 2022). Moreover, no city-wide, statistically grounded assessment exists that examines whether poorer wards are systematically hotter or have experienced greater warming over the past decade. While reproducible LST workflows have been demonstrated in other cities (Mohiuddin & Mund, 2024), such approaches have not been applied to Dhaka in combination with poverty indicators.

This lack of integrated, spatially explicit evidence limits the ability of planners and policymakers to identify high-risk communities and design climate-equitable adaptation strategies. Addressing this gap is essential for understanding Dhaka's heat–poverty nexus and supporting targeted interventions in vulnerable areas.

1.4 Research Questions

The central research question guiding this study is:

How do thermal hotspots in Dhaka correlate with ward-level poverty distribution in 2017, and have wards with higher poverty levels experienced greater thermal intensification between 2014 and 2024?

1. How have annual and seasonal LST patterns evolved across Dhaka's wards between 2014 and 2024?
2. Where are the city's persistent ward-level thermal hotspots and cold spots located?
3. How do 2017 ward-level poverty patterns correspond with the spatial distribution of thermal hotspots?
4. Are long-term LST trends (warming or cooling) systematically different in wards with higher poverty levels compared to less poor wards?

1.5 Objectives of the Study

The study aims to address the central research question by pursuing the following objectives:

Primary Objectives

1. To generate annual and seasonal Land Surface Temperature (LST) composites for Dhaka from 2014 to 2024 using Landsat 8/9 Level-2 data with cloud and shadow masking.
2. To quantify long-term spatial and temporal LST trends across Dhaka's wards using non-parametric statistical methods (Mann–Kendall test and Theil–Sen slope estimator).

3. To identify persistent thermal hotspots and cold spots through percentile-based classification of LST composites.
4. To examine the spatial relationship between thermal hotspots and ward-level poverty distribution in 2017 using zonal statistics and correlation analysis.

Secondary Objectives

1. To assess whether wards with higher poverty levels have experienced greater thermal intensification between 2014 and 2024.
2. To identify high-risk wards where long-term warming, hotspot concentration, and socioeconomic vulnerability converge.
3. To develop a reproducible, semi-automatic workflow for linking multi-year thermal dynamics with socioeconomic indicators in data-scarce urban environments.

1.6 Significance of the Study

This study provides the first integrated, ward-level assessment of Dhaka's thermal dynamics and socio-economic vulnerability. By combining multi-year satellite-derived Land Surface Temperature (LST) with a 2017 poverty indicator, it offers a reproducible framework for identifying neighbourhoods where high temperatures and socio-economic disadvantage overlap. Scientifically, the study advances understanding of thermal inequality in a rapidly urbanising megacity where the social dimensions of heat exposure remain underexplored.

Methodologically, the research adapts a semi-automatic Landsat time-series workflow and demonstrates how Earth observation can be used to analyse heat–poverty interactions in data-scarce environments. The integration of LST retrieval, seasonal compositing, trend analysis, and ward-level socio-economic data within a single workflow provides a transferable approach for future urban climate studies.

From a policy perspective, the findings support climate-equitable planning by highlighting wards that would benefit most from targeted interventions such as urban greening, cool roofing, and heat-health measures. The results offer spatially explicit evidence that can inform Dhaka's Climate Action Plans and Bangladesh's National Adaptation Plan, contributing to more effective and socially just climate-adaptation strategies.

1.7 Report Structure

This report is organised into nine chapters. Chapter 1 introduces the study by outlining the background, problem statement, research gap, research question, objectives, and significance. Chapter 2 reviews the existing literature on Land Surface Temperature (LST), Urban Heat Island (UHI) dynamics, land-cover change, heat vulnerability, and thermal inequality. Chapter 3 presents the conceptual framework that guides the analytical approach. Chapter 4 describes the study area and the datasets used, including satellite imagery and ward-level socio-economic indicators. Chapter 5 details the methodological workflow, covering LST retrieval, composite generation, trend analysis, hotspot detection, and the integration of poverty data. Chapter 6 presents the empirical results, including spatial and temporal LST patterns, hotspot dynamics, poverty distribution, and heat–poverty interactions. Chapter 7 discusses these findings in relation to urban development, inequality, and climate-justice perspectives. Chapter 8 outlines

the limitations of the study. Chapter 9 concludes the report by summarising the key insights, highlighting policy implications, and suggesting directions for future research.

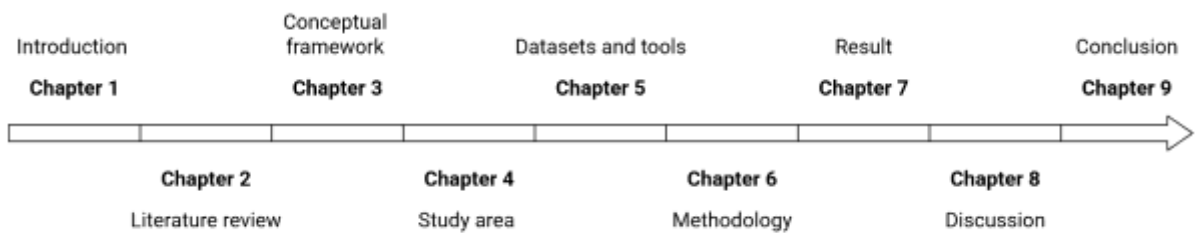


Figure 1.1 — Study Workflow Overview

Source: Own illustration by Mahbubul Alam (2026)

2. Literature Review

This chapter synthesises existing research on Urban Heat Islands, Land Surface Temperature dynamics, land-cover change, and heat-related vulnerability to establish the scientific foundation for the present study. To ensure a rigorous and transparent assessment of the literature, a systematic review process was applied, beginning with 250 records identified across major academic databases and narrowing to 30 studies that met all spatial, thermal, and socioeconomic relevance criteria (**Figure 2.1**). The selected studies are reviewed thematically to highlight what is known about Dhaka’s thermal environment, how land-surface processes shape heat patterns, and how vulnerability intersects with socioeconomic conditions. This structured approach provides the basis for identifying unresolved gaps and motivates the study’s focus on long-term thermal inequality.

2.1 Urban Heat Islands and LST Dynamics in Dhaka

A substantial body of research has examined the thermal behaviour of Dhaka and other South Asian cities, consistently showing that rapid urbanisation has intensified the Urban Heat Island (UHI) effect. Studies using Landsat, MODIS, and meteorological data report rising surface temperatures, expanding hotspot zones, and seasonal variations in UHI intensity (Uddin et al., 2024; Dewan et al., 2021; Jiang et al., 2024). Hotspot analyses using spatial statistics such as Getis-Ord Gi* further reveal that high-temperature clusters are concentrated in dense built-up areas (Hussain et al., 2023). Evidence from Dhaka also indicates that informal settlements experience significantly higher LST than formal neighbourhoods, suggesting early signs of thermal inequality (Hossain et al., 2022). Together, these studies provide a strong baseline for understanding Dhaka’s thermal environment but focus primarily on physical patterns rather than social disparities.

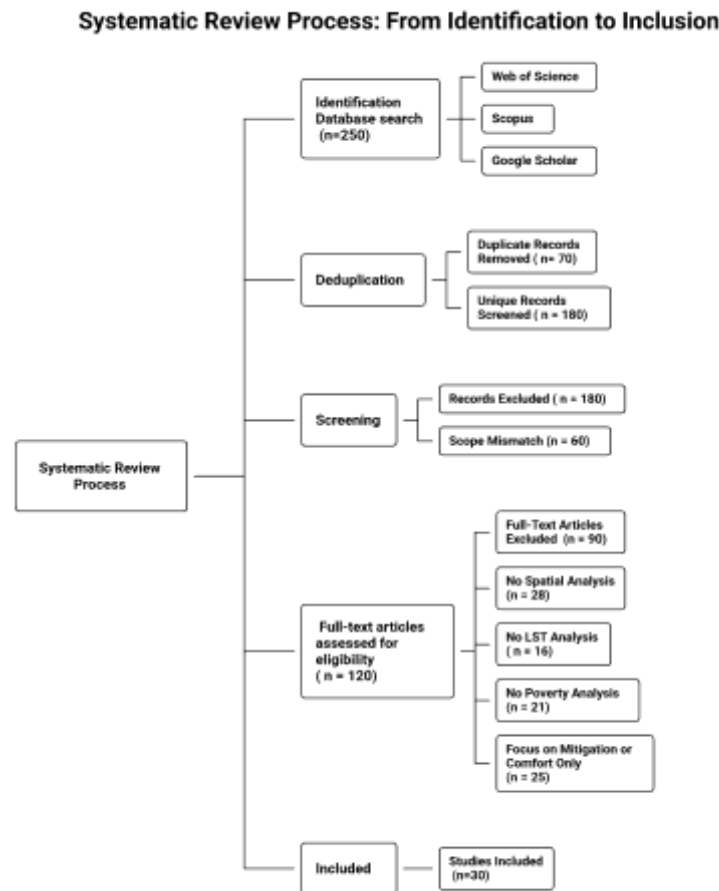


Figure 2.1 — Systematic Review Flowchart

Source: Own illustration based on PRISMA guidelines (2026).

2.2 Land-Cover Change and Its Influence on Surface Temperature

The relationship between land-cover change and LST has been widely documented in Dhaka and other tropical megacities. Research consistently shows that increases in built-up area and decreases in vegetation lead to higher surface temperatures (Trotter et al., 2017; Imran et al., 2021; Faisal et al., 2022). Historical analyses reveal that Dhaka's built-up area has more than doubled over recent decades, accompanied by a 3–5°C rise in LST (Dewan et al., 2009; Faisal et al., 2022). Predictive modelling studies using CA-ANN approaches further suggest that LST will continue to rise if current land-use trends persist (Kafy et al., 2021; Ahmed et al., 2013). These findings highlight the strong biophysical drivers of urban heat but do not address how these changes intersect with socioeconomic vulnerability.

2.3 Heat Vulnerability, Poverty, and Public Health

A parallel body of literature emphasises that heat exposure is not only an environmental issue but also a social one. Studies in Bangladesh and globally show that low-income populations face greater heat-related health risks due to poor housing quality, occupational exposure, and limited access to cooling resources (Nahar et al., 2010; Kjellstrom et al., 2009). Vulnerability

assessments in Dhaka identify several high-risk thanas where socioeconomic disadvantage, population density, and inadequate infrastructure amplify heat stress (Talukdar et al., 2022; Ryhan et al., 2023). Broader poverty studies in Bangladesh further demonstrate that socioeconomic vulnerability is closely linked to environmental risk exposure (Akter et al., 2013). Despite this evidence, no study has systematically examined whether Dhaka's poorer wards are also its hottest or whether they have warmed more rapidly over time.

2.4 LST Retrieval Methods and Time-Series Approaches

Methodological studies provide the technical foundation for analysing LST. Global reviews outline the strengths and limitations of different retrieval algorithms (Li et al., 2013), while validation studies in Dhaka compare algorithm performance under local conditions (Mallick et al., 2022). Recent advances in cloud-based platforms such as Google Earth Engine have enabled reproducible, semi-automatic workflows for generating multi-year LST time series (Mohiuddin & Mund, 2024). These workflows allow consistent processing of large satellite archives and support long-term trend analysis. However, such methods have not yet been applied to Dhaka in combination with socioeconomic indicators, leaving a methodological gap in understanding thermal inequality.

2.5 Mitigation Strategies and Policy Context

Research on mitigation strategies highlights the potential of greening interventions to reduce urban temperatures. Field experiments and modelling studies in Dhaka demonstrate that green roofs and vegetated surfaces can significantly lower surface temperatures (Alam et al., 2021; Rashid et al., 2017). Global meta-analyses confirm the cooling benefits of urban greening (Bowler et al., 2010), while regional studies identify threshold vegetation levels needed to achieve meaningful cooling (Uddin et al., 2022). Policy documents such as Bangladesh's National Adaptation Plan (2023) and Dhaka's Climate Action Plan (2024) emphasise heat resilience but lack spatially explicit evidence on which communities face the greatest thermal burden. This limits the effectiveness of targeted adaptation strategies.

2.6 Synthesis and Justification for the Present Study

The reviewed literature provides strong evidence on Dhaka's rising temperatures, land-cover change, and heat-related vulnerability. However, several critical gaps remain. Most studies rely on limited satellite dates, focus exclusively on physical variables, or do not integrate socioeconomic indicators. No research has yet combined multi-year LST analysis with ward-level poverty data to examine whether poorer areas are systematically hotter or have experienced greater thermal intensification. Furthermore, reproducible LST time-series workflows exist but have not been applied in a socially informed context.

This study addresses these gaps by integrating a decade of Landsat-derived LST with a 2017 poverty index to examine the spatial and temporal dimensions of thermal inequality in Dhaka. By linking environmental and socioeconomic indicators, the research contributes to a more comprehensive understanding of urban heat risk and supports climate-equitable planning.

2.7 Research Gap

A systematic review of 30 studies on LST, UHI dynamics, land-cover change, and urban vulnerability (**Figure 2.2**) in Dhaka and comparable cities reveals several critical gaps:

1. **Missing heat–poverty linkage** in Dhaka While multiple studies quantify LST trends and UHI intensity, none explicitly combine thermal data with a spatial poverty index for Dhaka. As a result, poverty and thermal exposure are typically analysed in isolation rather than as interacting dimensions of urban vulnerability (Dewan et al., 2021; Imran et al., 2021; Talukdar et al., 2022).
2. **Limited use of long-term** LST time series Many UHI studies rely on only three to five satellite images spread over decades, or focus on a single year or season, restricting insight into interannual variability and long-term thermal trajectories (Trotter et al., 2017; Sresto et al., 2022). In contrast, recent work in Phnom Penh demonstrates the value of dense Landsat time series for analysing seasonal and long-term thermal dynamics (Mohiuddin & Mund, 2024).
3. **Underexplored thermal inequality** Neighbourhood-scale evidence suggests that low-income and informal settlements experience higher LST and poorer thermal comfort (Hossain et al., 2022; Imran et al., 2021). However, no city-wide, statistically grounded analysis exists on how thermal hotspots spatially align with poverty.
4. **Lack of reproducible, semi-automatic** workflows for Dhaka Although semi-automatic workflows for LST time-series processing have been proposed elsewhere (Mohiuddin & Mund, 2024), Dhaka lacks a reproducible, transparent pipeline that integrates radiometric calibration, emissivity estimation, composite generation, and socioeconomic linkage.
5. **Insufficient integration with policy-relevant** vulnerability frameworks Recent heat-vulnerability indices for Dhaka (Talukdar et al., 2022; Ryhan et al., 2023) rely on coarse administrative units and indirect proxies, and do not incorporate pixel-level, multi-year LST data, limiting their policy relevance.

This study is designed to address these gaps by integrating multi-year LST analysis with ward-level poverty data to explore the spatial and temporal dimensions of thermal inequality in Dhaka.

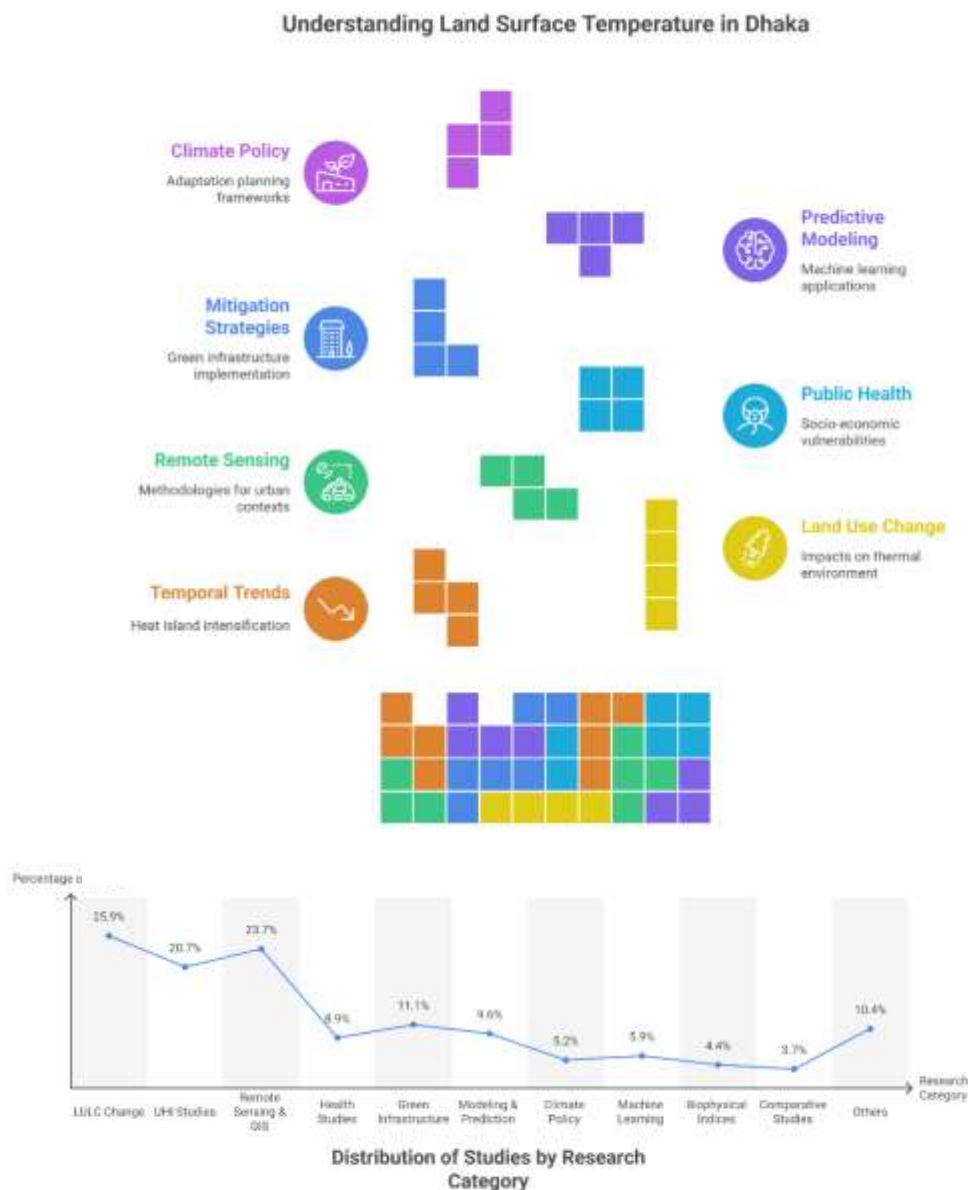


Figure2.2: Thematic Distribution of LST/UHI Research

Source: Own compilation from systematic review dataset (2026).

3. Conceptual Framework

The conceptual framework for this study is built on the understanding that urban heat emerges from the interaction between biophysical processes and socioeconomic conditions. In rapidly urbanising cities such as Dhaka, land-cover transformation—particularly the expansion of impervious surfaces and the reduction of vegetation—modifies the surface energy balance and contributes to elevated Land Surface Temperature (LST) (Imran et al., 2021; Faisal et al., 2022). Built-up intensity, vegetation cover, and surface materials are widely recognised as key determinants of spatial temperature variation, making LST a reliable indicator of urban thermal stress (Dewan et al., 2021; Hossain et al., 2022).

Heat exposure, however, is not distributed evenly across urban populations. International research shows that low-income communities often reside in densely built, poorly ventilated areas with limited vegetation, conditions that intensify thermal stress and reduce adaptive capacity (Chakraborty et al., 2019; Singh et al., 2020). Evidence from Dhaka reflects similar patterns, with informal settlements and socio-economically disadvantaged wards experiencing higher temperatures and greater vulnerability (Hossain et al., 2022; Imran et al., 2021). This highlights that urban heat is both an environmental and a social phenomenon, shaped by inequalities in housing quality, infrastructure, and access to cooling resources.

Building on these insights, the framework integrates three analytical components. First, multi-year LST derived from both Landsat and MODIS is used to characterise spatial and temporal thermal patterns. Landsat provides fine-scale spatial detail for mapping hotspots and seasonal behaviour, while MODIS offers temporally dense observations that support long-term trend detection and cross-validation (Mohiuddin & Mund, 2024). Second, land-cover characteristics such as vegetation, built-up density, and surface moisture help explain the physical drivers of these thermal patterns (Faisal et al., 2022). Third, ward-level poverty data are incorporated to assess whether socio-economically disadvantaged areas correspond to higher temperatures, persistent hotspots, or stronger warming trajectories (World Bank & BBS, 2019).

Together, these components form a coherent pathway for assessing thermal inequality in Dhaka. The framework assumes that if heat exposure is socially differentiated, it will be observable through the spatial overlap between high LST and high poverty, as well as through differences in long-term thermal intensification between poorer and less poor zones. This conceptual structure guides the methodological design of the study and provides a basis for identifying communities most at risk from rising urban temperatures.

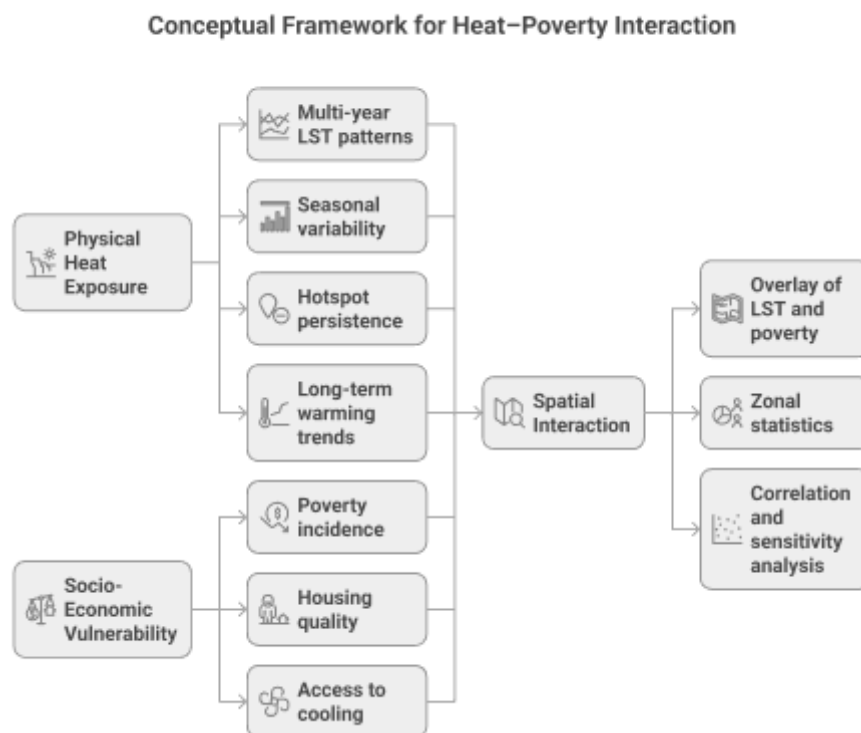


Figure 3.1: *Conceptual Framework for Heat–Poverty Interaction*Source:

Own conceptual model based on Dewan et al. (2021), Hossain et al. (2022), and Mohiuddin & Mund (2024).

This framework recognises that heat exposure and socio-economic vulnerability are mutually reinforcing. Poorer communities often live in hotter areas and have fewer resources to adapt, creating a cycle of thermal inequality. Most studies rely on a few satellite scenes per decade, limiting insight into long-term warming trajectories.

4. Study Area

4.1 Geographic Setting of Dhaka

Dhaka is located in the central part of Bangladesh within the lower Ganges–Brahmaputra–Meghna delta system. The metropolitan region has expanded rapidly over recent decades, forming the core of the Greater Dhaka Megalopolis. Historical analyses show that built-up areas more than doubled between 1989 and 2005, reflecting intense urban growth and land conversion (Dewan et al., 2009). The city is bounded by major rivers such as the Buriganga, Turag, and Balu, which shape its hydrological and environmental conditions.

4.2 Climate and Seasonal Characteristics

Dhaka experiences a humid subtropical to tropical monsoon climate with strong seasonal contrasts. Meteorological observations show that UHI intensity varies seasonally, with the strongest morning UHI occurring in winter (Jiang et al., 2024). Long-term climate projections also indicate rising temperatures across Bangladesh, with increases of 1.1–4.8°C expected by 2100 (Caesar et al., 2015). These climatic patterns interact with rapid urbanization to intensify local heat exposure.

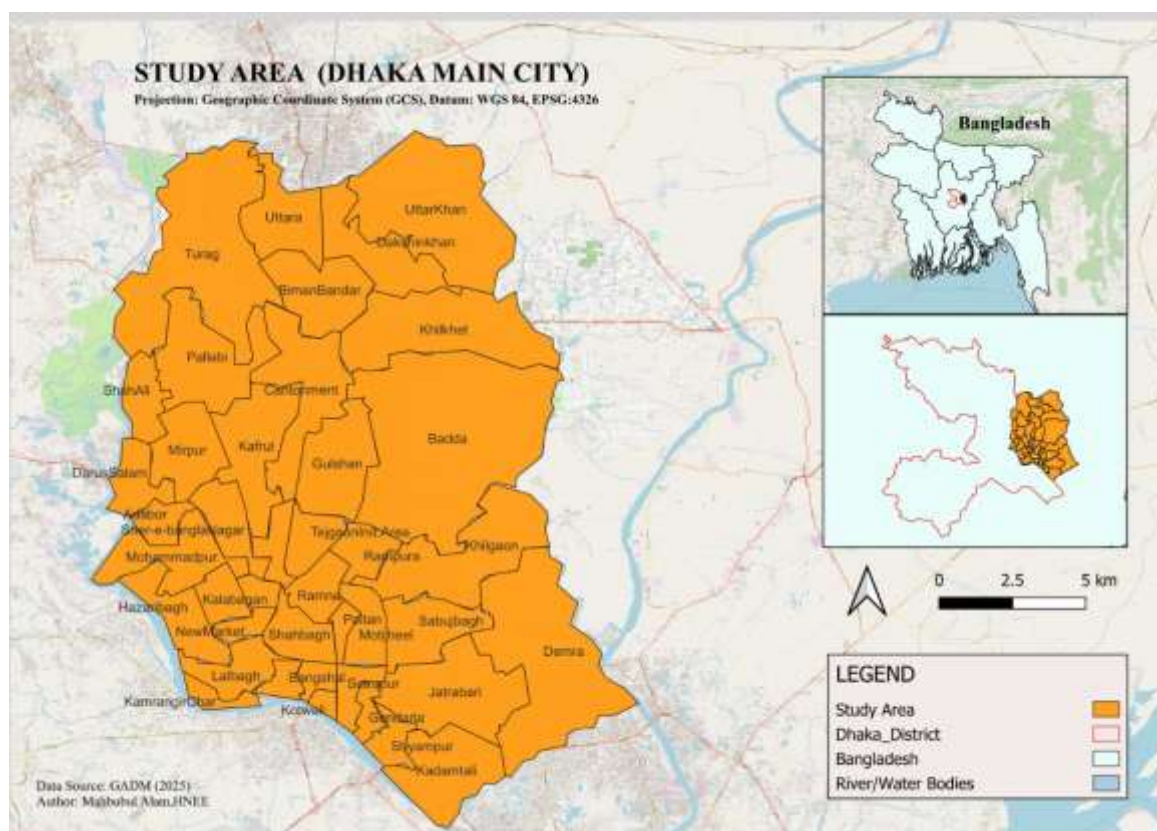


Figure 4.1: Map of Dhaka Metropolitan Area

Source: GADM administrative boundaries (2023); Basemap (USGS, 2024). Map created by Mahbubul Alam (2025).

4.3 Urbanization Trends and Land Cover Change

Dhaka is one of the fastest-urbanising megacities in South Asia, and multi-temporal remote sensing studies consistently document rapid and extensive land-cover transformation. Built-up area expanded by approximately 67% between 1993 and 2020 (Imran et al., 2021), accompanied by a 3–5°C rise in Land Surface Temperature over the same period, largely driven by the loss of vegetation and wetlands (Faisal et al., 2022). Earlier assessments show that the city’s urban growth has followed an unsustainable trajectory since the 1990s (Dewan et al., 2009). Seasonal variations in LST are strongly associated with land-cover indicators such as NDVI and NDBI, highlighting the sensitivity of surface temperature to vegetation and built-up intensity (Sresto et al., 2022). Predictive modelling further suggests continued warming and expansion of impervious surfaces by 2030 if current trends persist (Kafy et al., 2021; Ahmed et al., 2013). Regional analyses reinforce these findings, showing that Dhaka exhibits one of the strongest Surface Urban Heat Island signals in South Asia (Hassan et al., 2021).

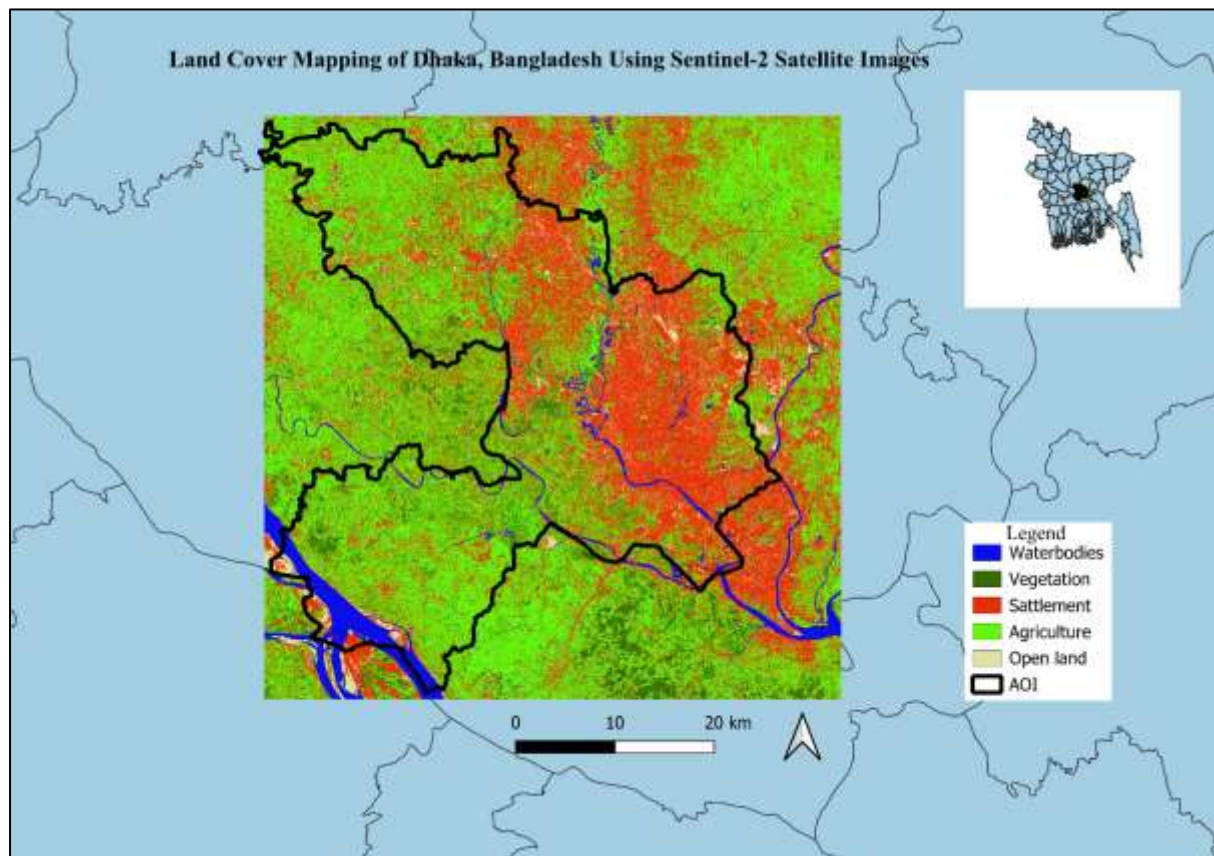


Figure 4.2: Land Cover Mapping of Dhaka, Bangladesh.

Based on Sentinel-2 MSI image (MGRS tile 45QXD, acquisition date 14 February 2025, cloud cover 5%, processing level Level-1C), classified using the Minimum Distance algorithm. Source: Copernicus Open Access Hub. Map created by Mahbubul Alam, 2025.

4.4 Socio-Economic Profile and Poverty Distribution

Dhaka's socio-economic landscape is characterized by high population density, widespread informal settlements, and significant inequality. Neighbourhood-scale analyses show that LST is consistently higher in slum areas due to poor housing materials, limited ventilation, and lack of vegetation (Hossain et al., 2022). Poverty and vulnerability are closely linked, with low-income communities facing greater exposure to heat stress and reduced adaptive capacity (Akter et al., 2013).

Heat-related health risks are well documented in Bangladesh. Occupational studies show that workers in informal sectors experience high levels of heat stress and illness (Nahar et al., 2010), while broader epidemiological research links temperature extremes to increased mortality (Hashizume et al., 2012; Burkart et al., 2022). These findings highlight the intersection of heat, poverty, and public health in Dhaka.

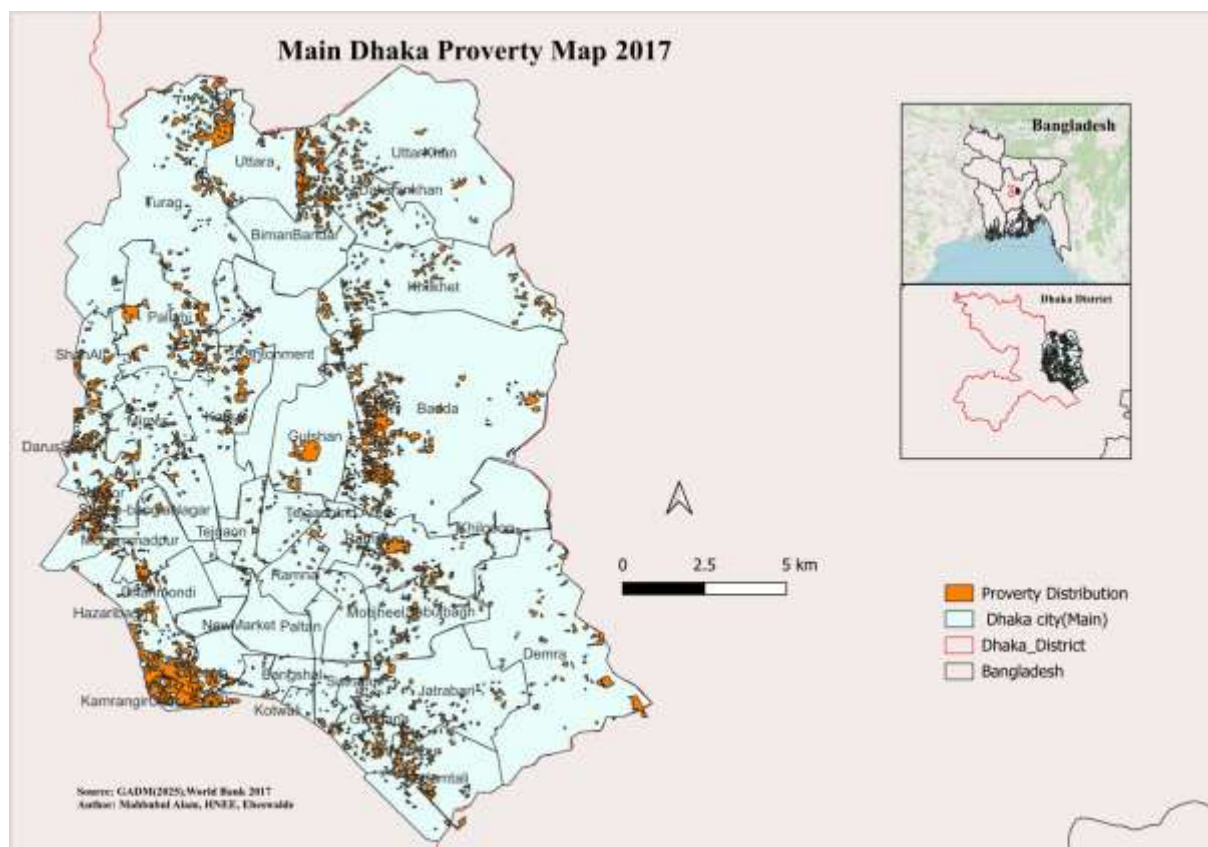


Figure 4.3: Dhaka poverty Map 2017

Source: EO4SD Urban Morphology Dataset (ESA, 2020). Processed by Mahbubul Alam (2025).

4.5 Rationale for Selecting Dhaka

Dhaka is selected as the study area due to its rapid and extensive land-cover transformation, which is among the fastest in South Asia and strongly linked to rising surface temperatures (Imran et al., 2021; Faisal et al., 2022). The city faces high climate vulnerability, experiencing recurrent heatwaves, flooding, and extreme weather events, with projections indicating continued warming in the coming decades (Caesar et al., 2015). Thermal inequality is also a growing concern, as low-income communities and informal settlements are disproportionately exposed to extreme heat due to poor housing conditions and limited access to cooling resources.

(Hossain et al., 2022; Nahar et al., 2010). Dhaka additionally benefits from extensive availability of high-quality remote sensing data, including long-term Landsat and MODIS archives and locally validated retrieval methods (Dewan et al., 2021; Mallick et al., 2022). Finally, the city holds strong policy relevance: national and municipal climate strategies increasingly prioritise heat resilience, yet lack fine-scale, spatially explicit evidence to guide equitable adaptation planning (Government of Bangladesh, 2023; Dhaka City Corporations, 2024). These factors collectively make Dhaka an ideal case for examining long-term thermal patterns and their intersection with socioeconomic vulnerability.

5. Data and Materials

5.1 Landsat 8/9 Thermal Data (2014–2024)

The study uses multi-temporal thermal infrared imagery from Landsat 8 OLI/TIRS and Landsat 9 OLI-2/TIRS-2 for 2014–2024. LST was derived from TIRS Band 10 (10.6–11.2 μm), originally 100 m and resampled to 30 m. Scenes with >60% cloud cover were excluded, and cloud, shadow, and cirrus pixels were removed using the QA_PIXEL mask in Google Earth Engine (GEE). Seasonal (pre-monsoon, monsoon, post-monsoon) and annual median composites were generated to reduce noise. A total of 399 cloud-free scenes were used, with improved coverage after Landsat 9 became operational in 2022.



Figure 5.1: Presents the temporal distribution of Landsat observations over Dhaka from 2014 to 2024.

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Processed by Mahbubul Alam (2025).

This study adapts the general concept of semi-automatic LST time-series processing demonstrated by Mohiuddin & Mund (2024), but extends it substantially by incorporating long-term trend analysis, hotspot detection, and socioeconomic vulnerability assessment for Dhaka. This ensures that the workflow is not only technically robust but also capable of addressing questions related to thermal inequality and climate justice.

5.2 MODIS LST Data (MOD11A2, 2014–2024)

MODIS MOD11A2 8-day composites (1 km) were used to complement Landsat’s spatial detail with long-term temporal consistency. Daytime and nighttime LST were extracted, and low-quality pixels were removed using QC_Day and QC_Night. MODIS supports trend detection, seasonal UHI assessment, and cross-validation of Landsat-derived LST.

5.3 Socioeconomic Data and Proxy Variables

5.3.1 Primary Poverty Dataset: EDSO 2017

The primary socioeconomic dataset used in this study is the EDSO 2017 poverty map, produced by the World Bank using small-area estimation techniques based on the Household Income and Expenditure Survey (HIES) 2016–17. The dataset provides ward-level poverty estimates by integrating survey observations with geospatial covariates, enabling a spatially continuous representation of poverty across Dhaka (World Bank, 2019). As no annual poverty dataset exists for the city, the 2017 poverty surface serves as a fixed socioeconomic baseline against which multiyear LST patterns (2014–2024) can be compared. Its spatial resolution and methodological transparency make it suitable for integration with pixel-based thermal indicators and for examining the spatial alignment between heat exposure and socioeconomic disadvantage.

5.3.2 Secondary Proxy: Building Density (NB_BUDENS)

To complement the static 2017 poverty baseline, building density (NB_BUDENS) was derived from a building-footprint shapefile of Dhaka and computed for each year from 2014 to 2024. High structural density is strongly associated with informal settlements and low-income neighbourhoods in Dhaka, where compact structures, limited ventilation, and minimal vegetation contribute to elevated thermal exposure. Using NB_BUDENS as a morphology-based socioeconomic proxy is therefore justified, as it provides an independent, spatially consistent indicator of structural vulnerability that can validate and cross-check the poverty patterns derived from EDSO 2017. This dual-indicator approach strengthens the robustness of the heat–poverty analysis by combining a statistically modelled poverty surface with a dynamic, morphology-based proxy.

5.3.3 Sensitivity Comparison of Poverty Indicators

Additional socioeconomic indicators (LIV_12, SEC_12, DW_ROOF) were reviewed but excluded because they are not available at ward level, limiting their suitability for citywide analysis. To understand how these alternative indicators behave relative to the selected datasets, a sensitivity comparison was conducted using the Heat–Poverty Vulnerability Index (HPVI). This comparison evaluates how different poverty proxies influence vulnerability classification and supports the methodological decision to prioritise EDSO 2017 while using NB_BUDENS as a validation proxy. The results of this assessment are presented in Figure X.

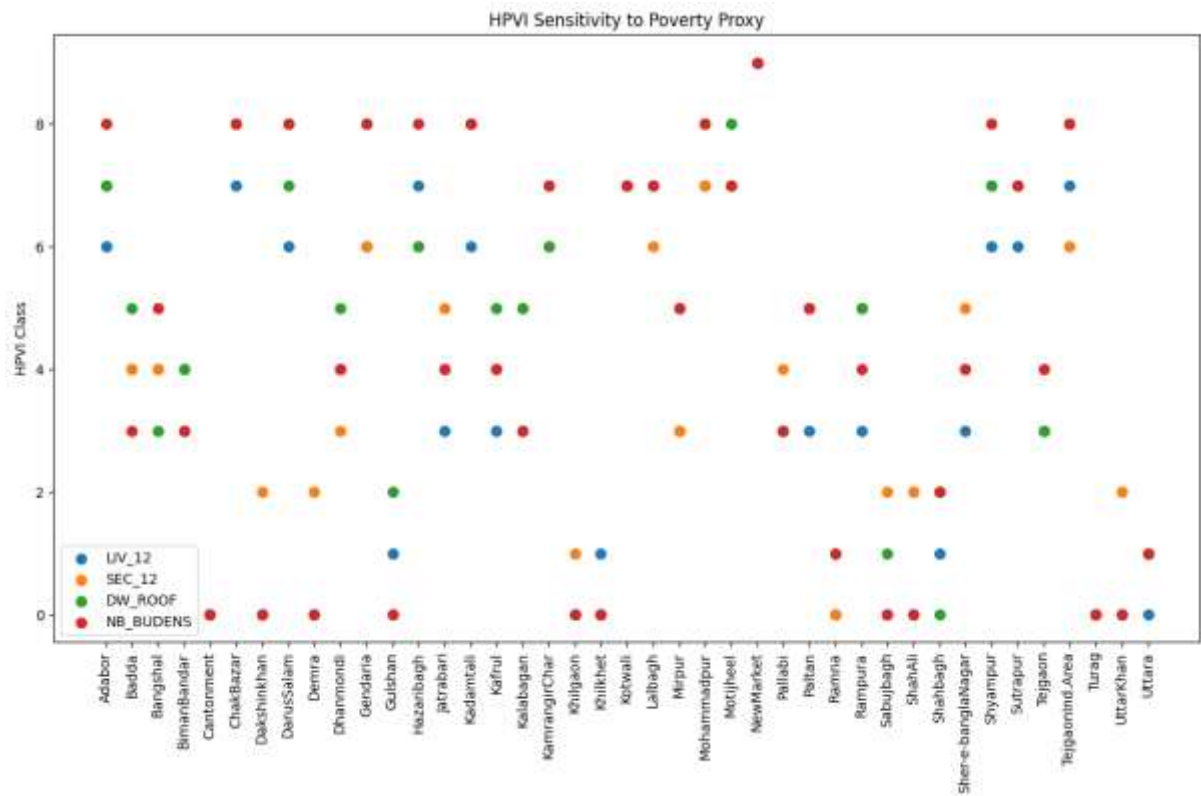


Figure 5.2. Sensitivity of Heat–Poverty Vulnerability Index (HPVI)

Source: Generated by the author using Python-based HPVI calculations (2026)

The scatter plot compares HPVI class values generated using four socioeconomic proxies (LIV_12, SEC_12, DW_ROOF, and NB_BUDENS) across Dhaka’s wards. The three survey-based indicators (LIV_12, SEC_12, DW_ROOF) show inconsistent and spatially unstable behaviour, reflecting their coarse resolution and limited suitability for ward-level analysis. In contrast, NB_BUDENS—derived from building-density patterns—shows the closest alignment with the reference poverty dataset (EDSO 2017), particularly in dense, low-income neighbourhoods. These results justify the selection of EDSO 2017 as the primary poverty dataset and support the use of NB_BUDENS as a robust morphology-based validation proxy.

5.4 Administrative Boundaries and Ancillary Data

Ward boundaries, roads, waterbodies, and other spatial layers were obtained from DNCC, DSCC, BBS, and OpenStreetMap. These datasets were used for masking, clipping, and spatial alignment of thermal and socioeconomic layers.

5.5 Software and Tools

Google Earth Engine (GEE) was used for LST retrieval, cloud masking, spectral index computation, and compositing. QGIS supported spatial editing and map production. Python was used for statistical analysis, trend estimation, and visualisation.

5.6 Spectral Indices (NDVI, NDBI, MNDWI)

NDVI, NDBI, and MNDWI were derived from Landsat OLI bands to characterise vegetation, built-up intensity, and surface moisture. These indices help interpret land-cover influences on LST.

1.NDVI — Vegetation Health

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

2.NDBI — Built-up Intensity

$$\text{NDBI} = \frac{\text{SWIR} - \text{NIR}}{\text{SWIR} + \text{NIR}}$$

3.MNDWI — Water and Moisture Detection

$$\text{MNDWI} = \frac{\text{Green} - \text{SWIR}}{\text{Green} + \text{SWIR}}$$

These indices help interpret land-cover patterns and explain spatial variations in LST.

5.7 Data Pre-Processing Summary

All datasets were harmonised through cloud filtering, LST retrieval, compositing, and normalisation. Spectral indices and building-density data were aligned with LST grids and aggregated to ward level. Zonal statistics produced the final dataset for analysing thermal exposure and socioeconomic vulnerability.

6. Methodology

6.1 Overview of the Workflow

The study follows an integrated geospatial workflow to quantify land surface temperature (LST), identify thermal hotspots, and examine their relationship with socioeconomic vulnerability in Dhaka. The workflow includes data acquisition (Landsat 8/9, MODIS, EO4SD building density, administrative boundaries), pre-processing (cloud masking, calibration, resampling, compositing), LST retrieval using the single-channel algorithm, seasonal and annual composite generation, long-term trend analysis, hotspot detection, and heat–poverty linkage assessment. This structured sequence ensures a reproducible and spatially coherent analysis.

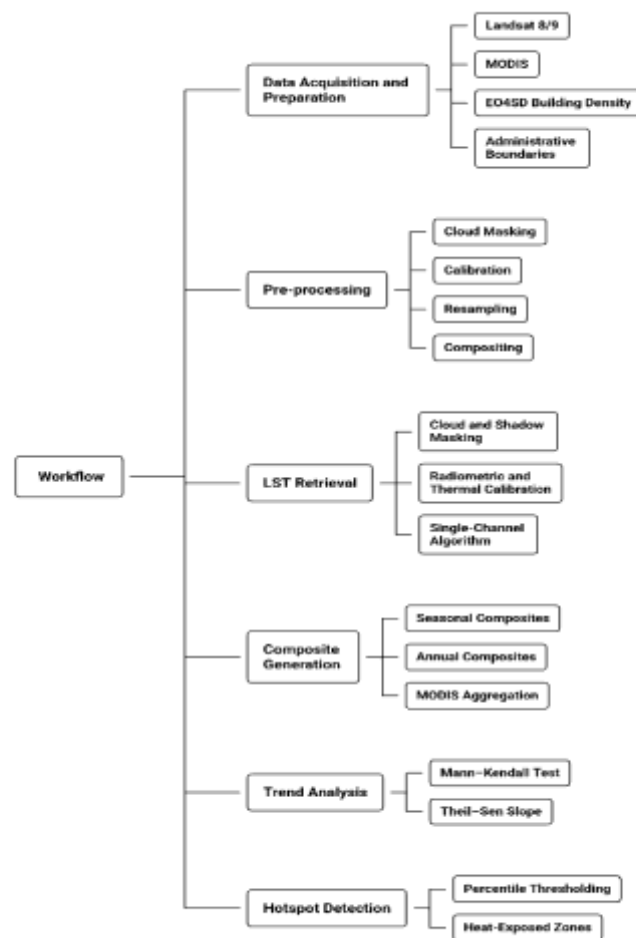


Figure 6.1: Processing Workflow Diagram

Source: Own illustration (2026).

6.2 LST Retrieval

Cloud and Shadow Masking

Given Dhaka's monsoon-driven cloud cover, scenes with more than 60% cloudiness were excluded. Cloud, shadow, and cirrus pixels were removed using the QA_PIXEL bitmask in Google Earth Engine (GEE), ensuring that only high-quality thermal observations were retained.

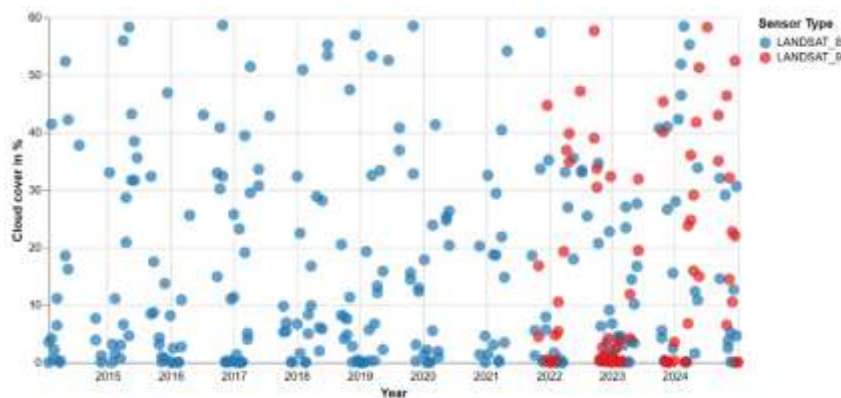


Figure 6.2: sensor wise images and their cloud percentage

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Processed by Mahbubul Alam (2025).

Radiometric and Thermal Calibration

Landsat digital numbers were converted to Top-of-Atmosphere radiance, followed by brightness temperature calculation using sensor calibration constants. Surface emissivity was estimated using NDVI-based emissivity models to account for differences between vegetation, built-up areas, and bare soil. These steps ensure that retrieved LST reflects true surface conditions.

Single-Channel Algorithm

LST was derived using the single-channel (SC) algorithm, which incorporates TOA radiance, atmospheric transmittance, upwelling and downwelling radiance, and surface emissivity. This method is widely validated for Landsat TIRS Band 10 and performs reliably in heterogeneous urban environments.

6.3 Annual and Seasonal LST Composite Generation

Seasonal LST composites were produced for pre-monsoon, monsoon, and post-monsoon periods using median pixel values to reduce noise and residual cloud effects. Annual composites were generated by aggregating all cloud-free scenes within each year. MODIS 8-day LST data were averaged annually to support long-term trend analysis from 2014 to 2024.

6.4 Trend Analysis

Mann–Kendall Test

The Mann–Kendall (MK) test was applied to detect monotonic trends in the land surface temperature (LST) time series. As a non-parametric and outlier-resistant method, it is well suited for environmental datasets that do not necessarily follow a normal distribution. The MK test evaluates whether a variable exhibits a statistically significant increasing or decreasing trend over time without requiring assumptions about data linearity or variance homogeneity (Mann, 1945; Kendall, 1975).

Mann–Kendall Test and Theil–Sen Estimator for LST Trend Analysis

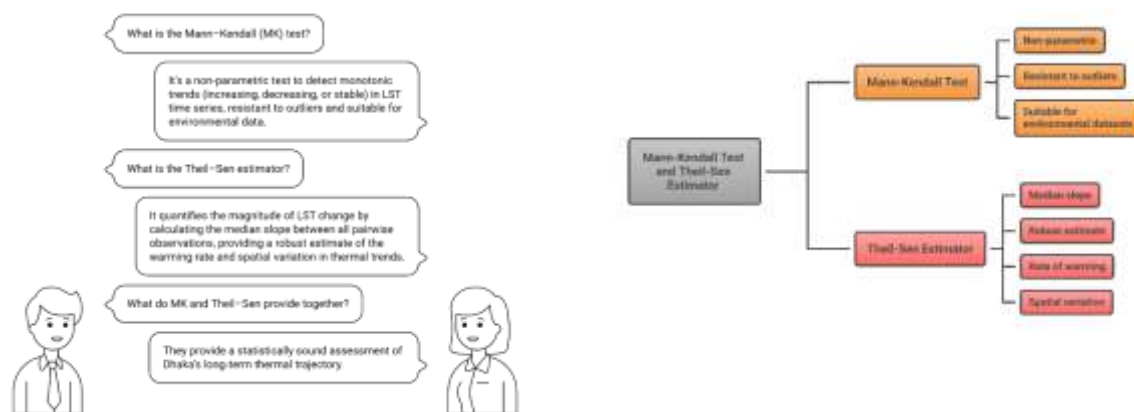


Figure 6.3 — Trend Analysis Workflow MK + Theil–Sen. (own presentation)

Theil–Sen Slope Estimator

The Theil–Sen estimator was used to quantify the magnitude of LST change by calculating the median slope among all pairwise combinations of observations. This approach provides a robust estimate of long-term warming rates and is less sensitive to extreme values than ordinary least squares regression. When applied spatially, the Theil–Sen slope enables assessment of spatial variability in thermal trends across urban zones (Theil, 1950; Sen, 1968).

6.5 Hotspot and Cold-Spot Detection

Thermal hotspots and cold spots were identified using percentile thresholds. Pixels above the 90th percentile were classified as hotspots, while those below the 10th percentile were considered cold spots. A minimum LST threshold of 20°C was used to delineate heat-exposed zones. This approach captures extreme thermal conditions consistently across seasons and years.

6.6 Poverty–Heat Linkage

Zonal Statistics

Ward-level zonal statistics were computed by overlaying LST composites, spectral indices (NDVI, NDBI, MNDWI), and building density (NB_BUDENS). Extracted metrics included mean and maximum LST, hotspot area share, vegetation and built-up intensity, and building density values.

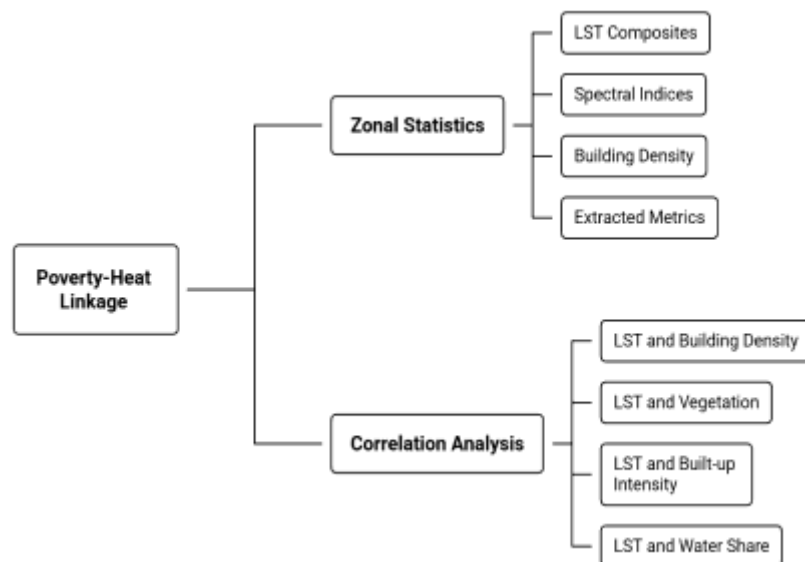


Figure 6.4: Heat–Poverty Linkage Workflow (own presentation)

6.7 Statistical Analysis Framework

To examine the relationships between surface temperature and key environmental and socioeconomic variables, a combination of correlation and group-comparison tests was applied. Pearson and Spearman correlation coefficients were used to quantify linear and monotonic associations between LST and explanatory variables including vegetation cover (NDVI), built-up intensity (NDBI), surface moisture (MNDWI), imperviousness (IBI), and building density (NB_BUDENS). These correlations provide a statistical basis for identifying the environmental factors most strongly associated with surface heat. To assess whether thermal conditions differ systematically across socioeconomic groups, two complementary tests were employed: an independent samples t-test to compare mean LST between high-poverty and low-poverty wards under approximate normality assumptions, and the non-parametric Mann–Whitney U test to provide a distribution-free validation of group differences. Together, these methods offer a robust analytical framework for evaluating both environmental drivers of heat and the presence of thermal inequality across Dhaka.

6.8 Ethical Considerations

The study relies exclusively on publicly available remote sensing and administrative datasets. No personal or household-level information was used. Building density served as a non-intrusive proxy for vulnerability. All data were processed solely for academic research, adhering to ethical standards for geospatial and socio-environmental studies.

6.9 Methodological Limitations

Several limitations should be acknowledged. The absence of ward-level poverty data required the use of building density as a proxy. Cloud contamination reduced the number of usable Landsat scenes, and minor uncertainties may arise from Landsat thermal calibration. MODIS's coarse spatial resolution (1 km) may smooth local variations, and the lack of ground-based temperature measurements limits validation. Spectral indices also cannot fully capture micro-scale urban morphology.

7.Result

This chapter presents the spatio-temporal dynamics of Land Surface Temperature (LST) in Dhaka from 2014 to 2024 and examines how thermal exposure intersects with socio-economic vulnerability. Only the essential figures are included in the main text to maintain clarity and narrative coherence. All supplementary scatterplots, diagnostic maps, and sensitivity analyses are provided in Appendix.

7.1 Temporal Patterns of Land Surface Temperature (2014–2024)

To characterise Dhaka's thermal environment, four complementary visualisations summarise the distribution, seasonal behaviour, annual variability, and long-term trajectory of Land Surface Temperature (LST) from 2014–2024. The boxplots show (Figure 6.1a) a stable thermal structure, with MinLST around 22 °C, MeanLST near 32 °C, and MaxLST exceeding 44 °C.

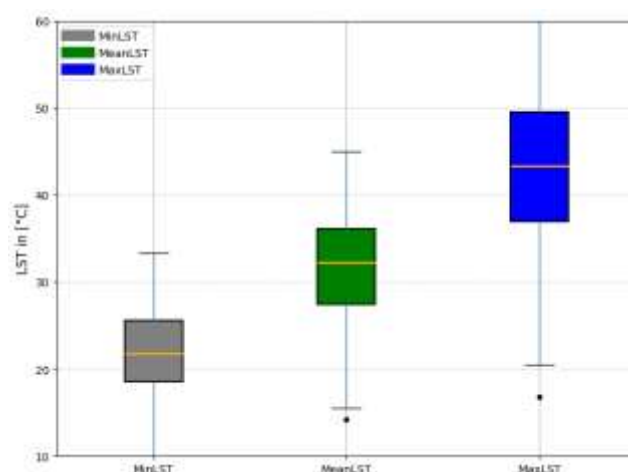


Figure 7.1a. Distribution of minimum, mean, and maximum LST (N = 399)

Figure 7.1b Monthly mean LST heatmap (2014–2024). Clear seasonal cycles appear, with cooler conditions in January–March and strong pre-monsoon heating from March–May.

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Processed by Mahbubul Alam (2025).

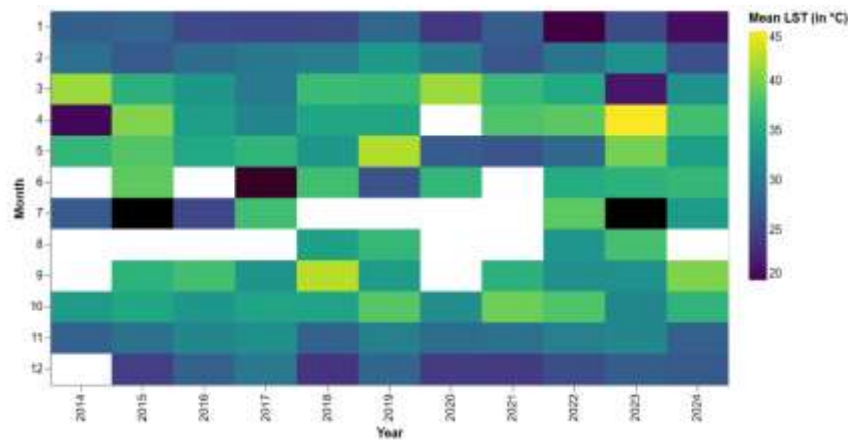


Figure 7.1b: Monthly Mean LST Heatmap (2014–2024)

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Processed by Mahbubul Alam (2025).

Missing data points are indicated by white cells. This visualization complements the boxplot by revealing temporal persistence and fluctuation in surface heating, critical for identifying climate-sensitive periods.

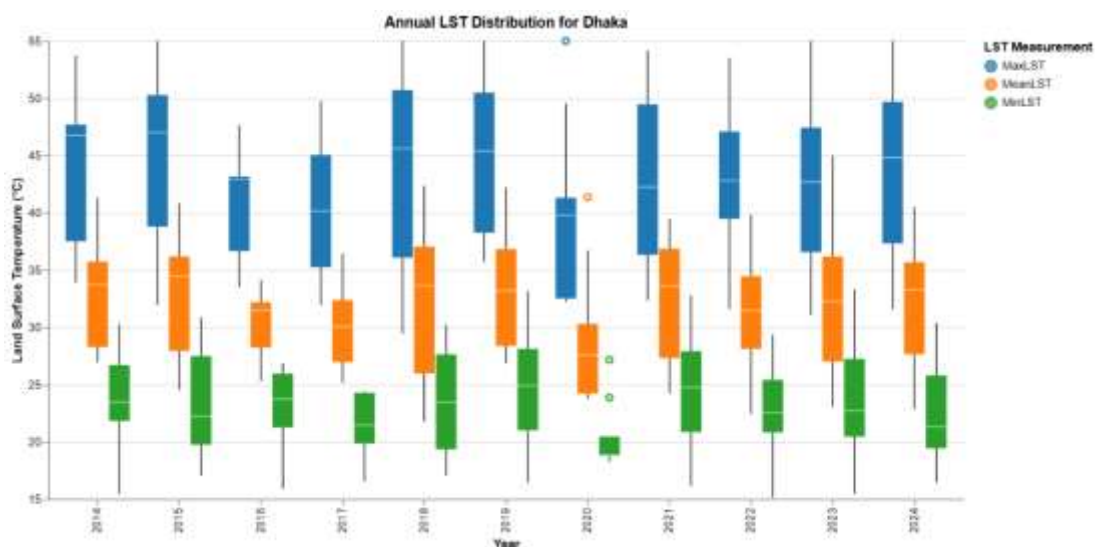


Figure 6.1c: Annual LST Distribution for Dhaka (2014–2024)

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Processed by Mahbubul Alam (2025).

This figure (Figure 7.1c) presents year-wise boxplots for MaxLST, MeanLST, and MinLST across Dhaka. Each year includes three color-coded boxes—blue for MaxLST, orange for MeanLST, and green for MinLST—allowing direct comparison of thermal spread over time. The median MaxLST values remain consistently high, often exceeding 45 °C, while MeanLST shows moderate variation and MinLST remains relatively stable. Outliers suggest episodic heat extremes. This visualization reinforces the presence of persistent urban heat stress and highlights years with elevated thermal exposure.

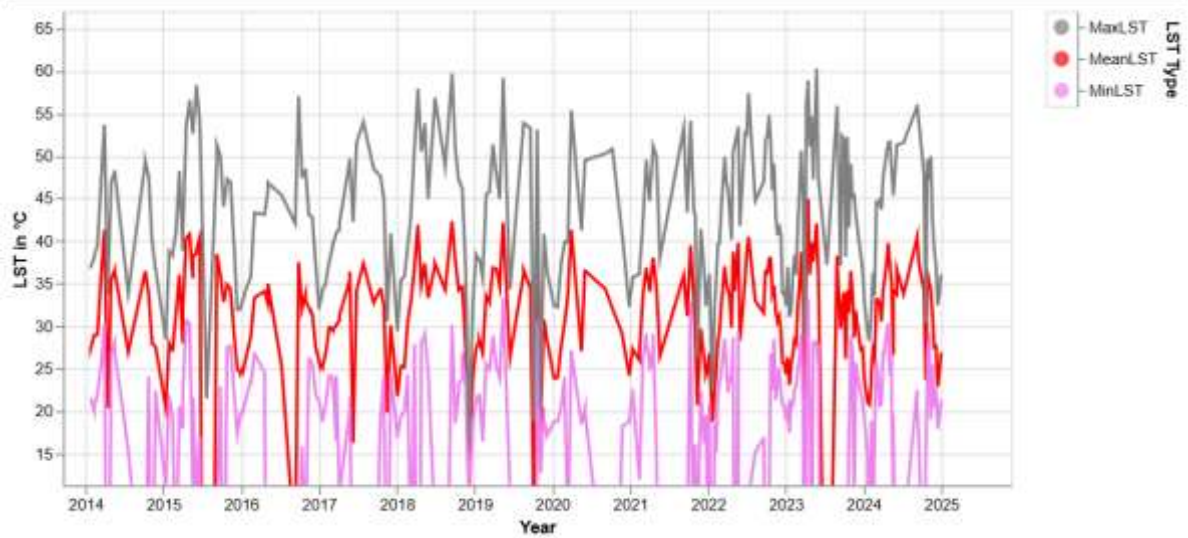


Figure 7.1d: Decadal LST Variation (2014–2024)

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Processed by Mahbulul Alam (2025).

This final Figure (7.1d) zooms in on the most recent decade, showing annual trends in MaxLST, MeanLST, and MinLST. MaxLST values frequently exceed 60 °C, while MeanLST fluctuates between 25 °C and 40 °C. MinLST remains mostly below 30 °C. The visualization highlights intra-annual variability and confirms the persistence of high-temperature episodes. The color-coded lines (gray for MaxLST, red for MeanLST, pink for MinLST) allow for intuitive tracking of each metric's trajectory.

Together, these four figures offer a comprehensive view of LST behavior—statistically, seasonally, annually, and historically—laying the foundation for vulnerability assessment and policy-relevant thermal zoning. The integration of distributional and temporal insights enables a nuanced understanding of heat exposure in Dhaka and supports targeted climate adaptation strategies.

7.2 Spatial and Seasonal Patterns of Land Surface Temperature (2017)

This section provides a detailed examination of Dhaka's Land Surface Temperature (LST) patterns for the year 2017, selected as a representative baseline due to its complete seasonal coverage and minimal cloud contamination. The analysis integrates monthly LST maps, seasonal transition curves, zone-wise temperature differences, and heat-risk classifications to capture both spatial variability and seasonal dynamics. By focusing on a single, data-rich year, this section highlights the structural thermal characteristics of Dhaka's urban landscape—identifying persistent hotspot zones, cooler peri-urban areas, and zones exhibiting significant seasonal temperature change (>5 °C). These insights form the foundation for the long-term trend analysis presented in Section 7.3.

Monthly LST Distribution

Figure 7.2a presents zone-wise monthly LST maps for January–May and October–December, using a unified colour scale (20–45 °C) to ensure comparability. June–September are excluded due to cloud contamination.

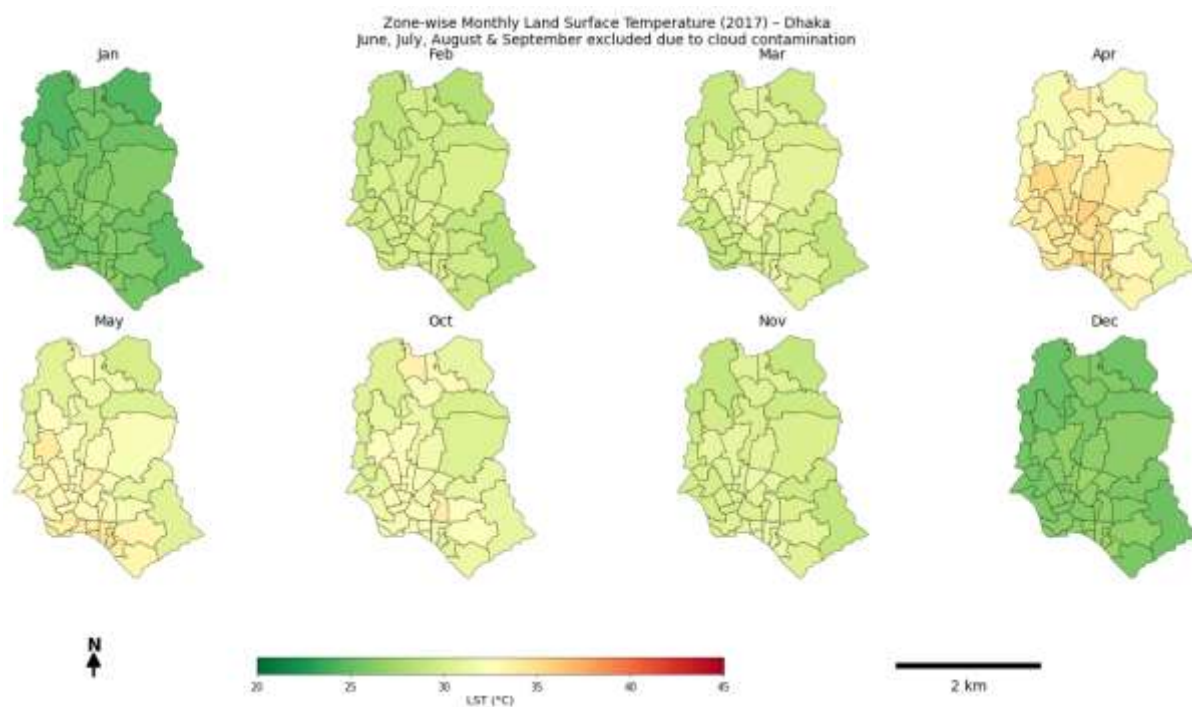


Figure 7.2a: Zone-wise Monthly LST Maps (2017)

Source: MOD11A2 V6 (NASA LP DAAC, 2014–2024). Trend analysis by Mahbubul Alam (2025).

The maps show strong pre-monsoon heating in March–May, particularly in dense urban cores, while November–January display widespread cooling, especially in vegetated and peri-urban zones. A $>5^{\circ}\text{C}$ seasonal-change threshold was applied to identify zones with meaningful thermal variability; only these zones are labelled in subsequent figures.

Seasonal LST Variation

Figure 7.2b illustrates seasonal LST variation across Dhaka, highlighting only zones that exceed the $>5^{\circ}\text{C}$ threshold. Hotspot and cool-spot zones are identified based on April (peak heating) and November (cooling) extremes. The figure captures the city's characteristic thermal rhythm: strong pre-monsoon heating, moderated monsoon temperatures, and a clear winter transition. Urban cores consistently show the highest seasonal variability.

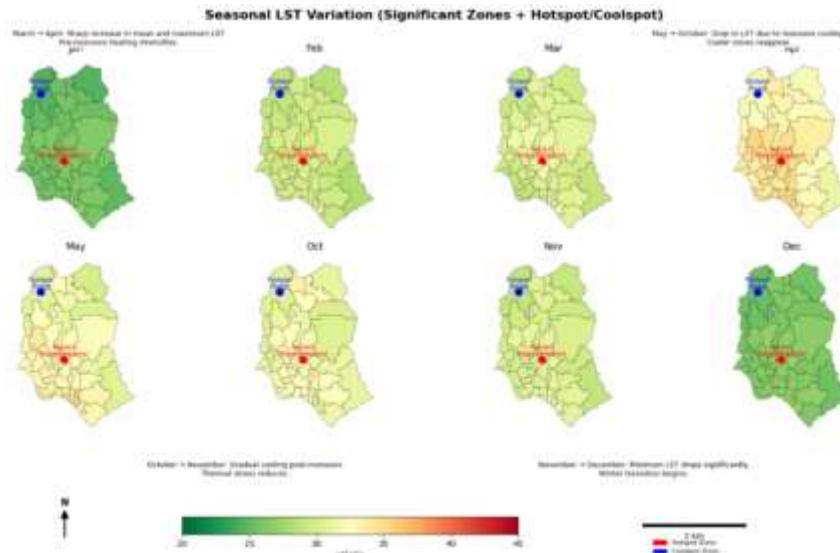


Figure 7.2b: Seasonal LST variation across Dhaka 2017

Source: MOD11A2 V6 (NASA LP DAAC, 2014–2024). Trend analysis by Mahbubul Alam (2025).

Seasonal Transitions and Extreme

Figure 7.2c shows monthly minimum, mean, and maximum LST values for 2017. Temperatures peak in April, with maximum LST reaching $\sim 36^\circ\text{C}$, and decline sharply by December. The shaded Min–Max range highlights seasonal amplitude, which is widest during the pre-monsoon period. These temporal patterns align closely with the spatial distribution of high-variability zones identified earlier.

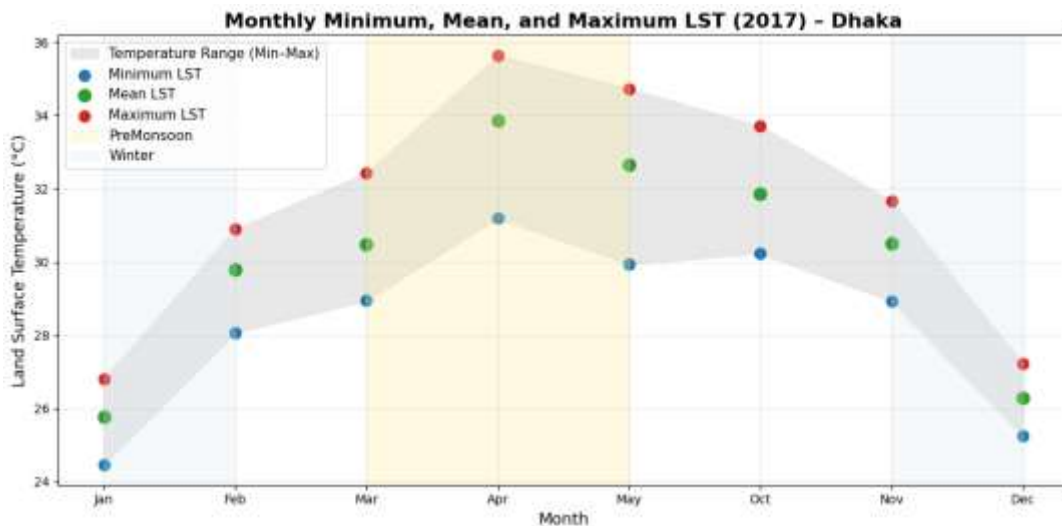


Figure 7.2c: Monthly Min/Mean/Max LST Trend Plot

Source: MOD11A2 V6 (NASA LP DAAC). Analysis based on 2017 data using GEE Python API. Mahbubul Alam (2025).

Zone-Wise Δ LST (April–December 2017)

Figure 7.2d presents the zone-wise LST difference between April and December (Table 1, Appendix 3). Most zones show positive Δ LST values, indicating substantial cooling from the pre-monsoon peak to winter. Urban cores such as Tejgaon and KamrangirChar exhibit Δ LST > 6 °C, reflecting strong thermal inertia and heat retention. Peri-urban and vegetated zones show lower Δ LST, indicating more stable thermal profiles. Table 1 (Appendix) lists all zones with Δ LST > 5 °C, confirming the robustness of the threshold-based classification.

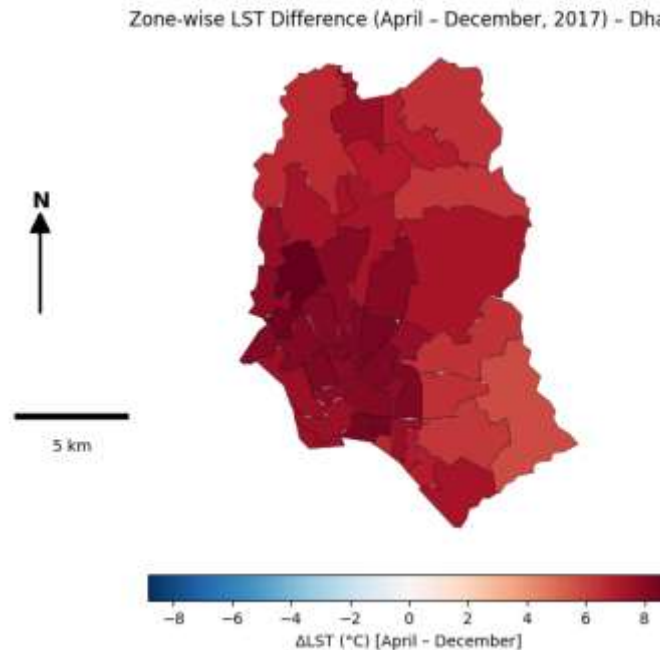


Figure 7.2d: Zone-Wise Δ LST (April–December)

Source: MOD11A2 V6 (NASA LP DAAC). Analysis based on 2017 data using GEE Python API. Mahbubul Alam (2025).

Supplementary Heat-Risk Classification

A supplementary K-means clustering analysis classified seasonal heat-risk zones using LST, NDVI, and CDI. The results show that pre-monsoon months carry the highest heat-risk levels, monsoon months moderate but do not eliminate hotspots, and winter months retain structural hotspots in dense urban cores. Full clustering outputs are provided in Appendix 1A.

7.3 Trend Analysis of Land Surface Temperature (2014–2024)

This section presents the long-term trend analysis of pre-monsoon Land Surface Temperature (LST) across Dhaka from 2014 to 2024 using two complementary non-parametric methods: the Mann–Kendall (MK) test for detecting monotonic trends and the Theil–Sen estimator for quantifying the rate of change (°C/year). Together, these methods provide a robust assessment of spatial and temporal thermal dynamics across the city’s 41 administrative zones.

Trend Significance (Mann–Kendall Test)

The MK test was applied to annual LST time series for all 41 zones to determine whether a statistically significant monotonic trend exists. The results show that the majority of zones do not exhibit a significant trend ($p \geq 0.05$), reflecting substantial interannual variability in Landsat-derived LST and the influence of cloud-related data gaps. Only a small cluster of dense inner-city zones—primarily in Old Dhaka—shows significant decreasing trends ($p < 0.05$). These include Shyampur, Gendaria, Kotwali, KamrangirChar, Dhanmondi, Mohammadpur, and ChakBazar. No zone shows a statistically significant increasing trend.

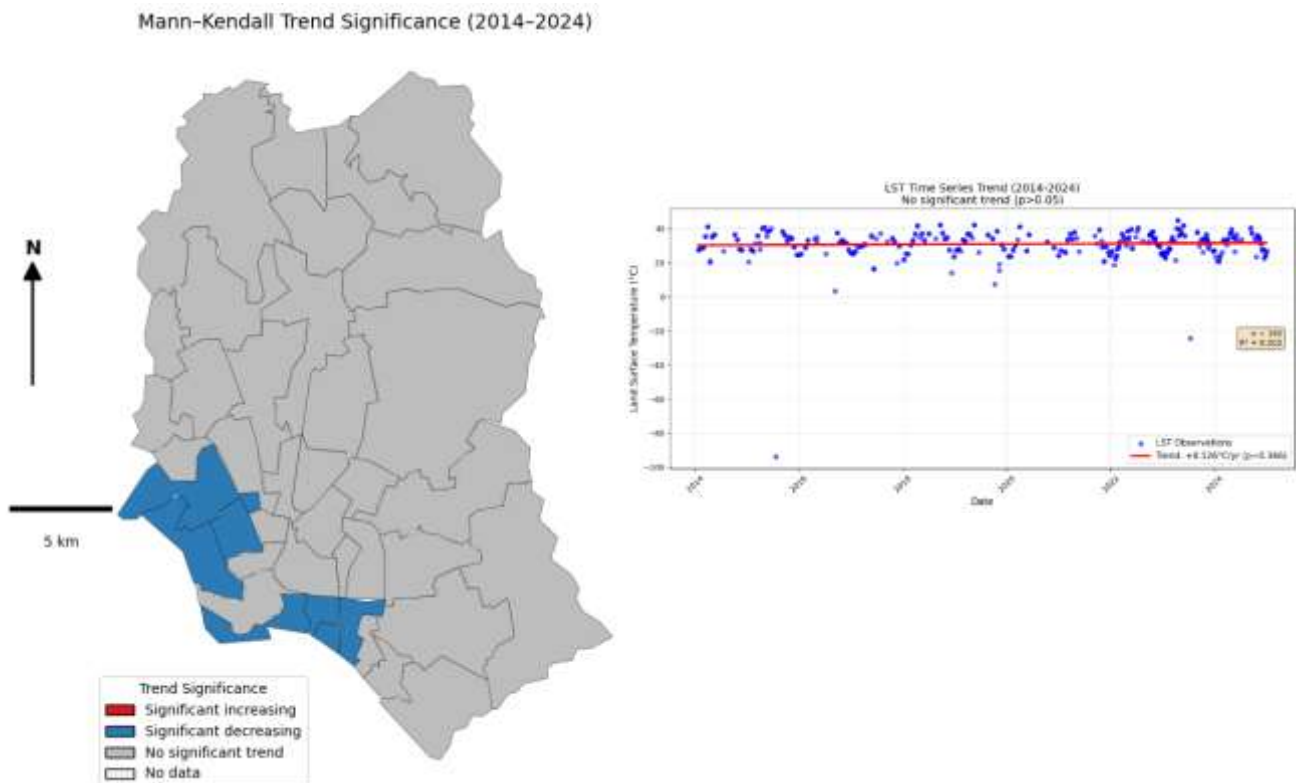


Figure 7.3a Mann–Kendall Trend Significance (2014–2024)

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Statistical analysis by Mahbulul Alam (2025).

This map displays the MK trend categories across Dhaka. Zones with significant decreasing trends ($p < 0.05$) are shown in blue, non-significant trends in grey, and significant increasing trends (if present) in red. The spatial pattern highlights that significant monotonic change is limited to a few inner-core zones, while most of the city shows no statistically detectable trend.

7.3.2 Magnitude of Change (Theil–Sen Slope)

The Theil–Sen estimator quantifies the rate of LST change in $^{\circ}\text{C}$ per year. The spatial pattern is dominated by negative slopes, indicating cooling or stable conditions across most of Dhaka. Slope values range from $+0.049^{\circ}\text{C}/\text{year}$, representing weak warming, to $-0.164^{\circ}\text{C}/\text{year}$, representing strong cooling. Only three peripheral zones—Biman Bandar, Demra, and Uttara—show weak warming. The strongest cooling occurs in ChakBazar, Mohammadpur, Dhanmondi, Hazaribagh, and KamrangirChar, all of which are characterised by dense built-up structures and narrow street canyons.

This map visualizes the rate of LST change across Dhaka using a diverging colour scale centered at zero. Red tones indicate warming, while blue tones represent cooling. The map clearly shows that cooling dominates the spatial pattern, with only a few peripheral zones exhibiting weak warming.

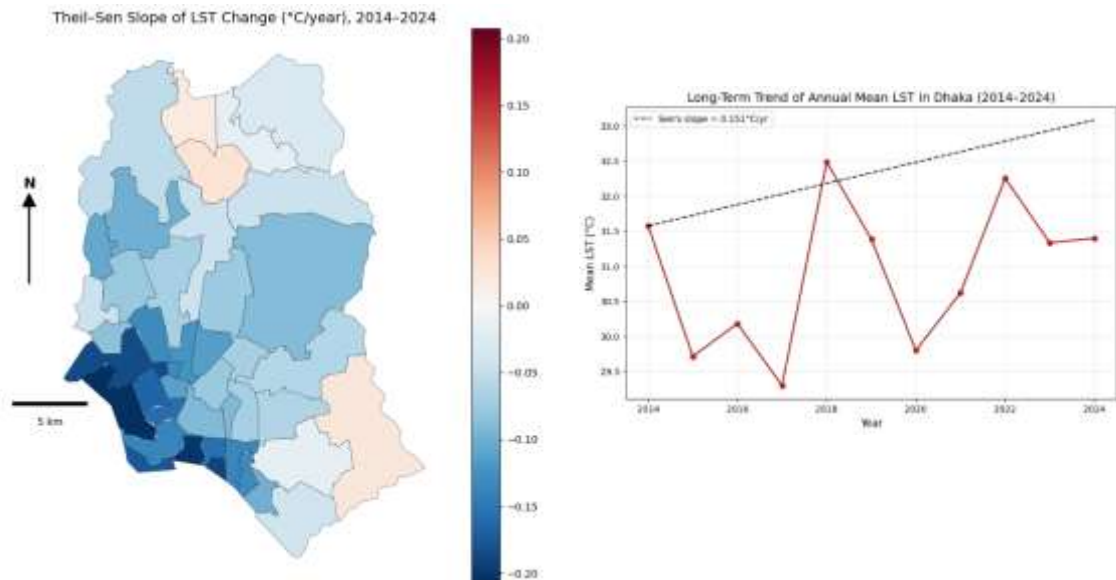


Figure 7.3b: Theil–Sen Slope of LST Change ($^{\circ}\text{C}/\text{year}$), 2014–2024

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Statistical analysis by Mahbubul Alam (2025).

7.3.3 Classified LST Trend Categories

To support clearer interpretation, the Theil–Sen slopes were grouped into six categories ranging from strong cooling to moderate warming. Most zones fall into the moderate or strong cooling classes, while warming is limited to a few northern and eastern peripheral areas. This classification highlights the spatial clustering of cooling in older, densely built neighbourhoods and the limited extent of warming elsewhere

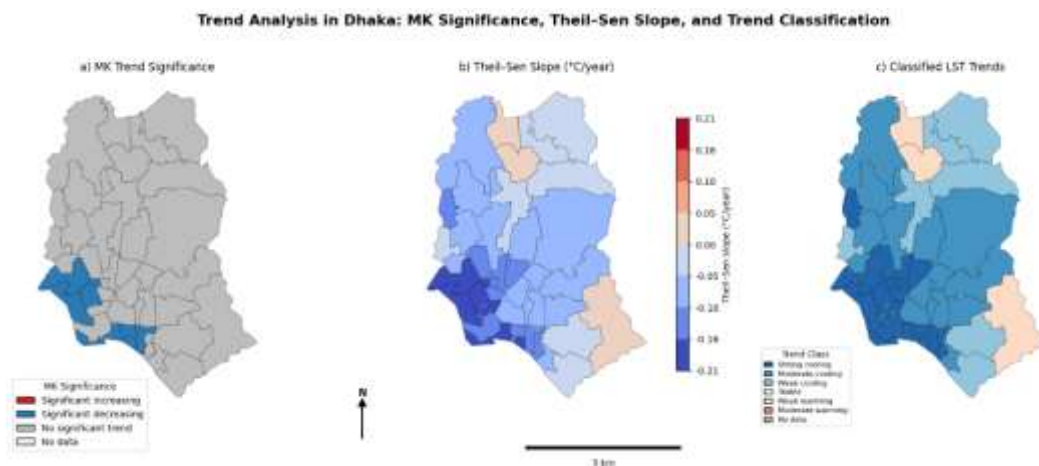


Figure 7.3c. Classified LST Trends Based on Theil–Sen Slope (2014–2024)

Source: Landsat 8/9 Collection 2 Level-2 (USGS, 2014–2024). Statistical analysis by Mahbubul Alam (2025).

This map categorizes each zone according to its long-term LST trend. Strong and moderate cooling dominate the central and western parts of Dhaka, while weak warming is limited to the northern and eastern periphery. The classification provides an intuitive summary of thermal change patterns.

7.3.4 Summary of MK and Theil–Sen Results

Taken together, the MK and Theil–Sen analyses indicate that Dhaka’s long-term thermal trajectory is characterised by predominantly negative trends, with cooling dominating most zones. Statistically significant trends occur only in the form of decreasing LST, and no zone shows significant warming. Weak warming is confined to a small number of peripheral zones with more open land and fewer data gaps. The strongest cooling signals appear in dense, older neighbourhoods, where urban morphology and atmospheric conditions may influence LST retrievals. These results provide a consistent statistical foundation for the subsequent analyses of hotspot persistence, vegetation–LST interactions, and the spatial relationship between heat exposure and socioeconomic vulnerability

These findings provide a robust statistical foundation for subsequent analyses in Sections 7.4 - 7.8 including hotspot persistence, heat–poverty interactions, and the development of the Urban Heat Vulnerability Index.

* MK and Theil–Sen data provided in Appendix 3, Table 2

7.4 Long-Term LST Trends and Urban Heat Expansion

This section examines how Dhaka’s thermal environment has evolved over the eleven-year period from 2014 to 2024. Building on the seasonal and annual patterns described in Sections 7.2 and 7.3, the analysis focuses on long-term warming trends, the spatial persistence of hotspots, and the expansion of heat-prone areas. Together, these indicators provide a comprehensive view of how urban growth and land-cover change have shaped Dhaka’s surface temperature regime.

Annual LST Patterns and Persistent Hotspots

The zone-wise annual LST maps (Figure 7.4a) reveal a stable but uneven thermal structure across Dhaka. Industrial and high-density residential zones—particularly Tejgaon, KamrangirChar, and Hazaribagh—consistently exhibit elevated temperatures throughout the decade. These areas combine dense built-up surfaces, limited vegetation, and high anthropogenic heat emissions, resulting in persistent hotspot conditions. In contrast, peri-urban zones along the northwest and southwest fringes remain comparatively cooler, reflecting the moderating influence of vegetation, open land, and lower development intensity.

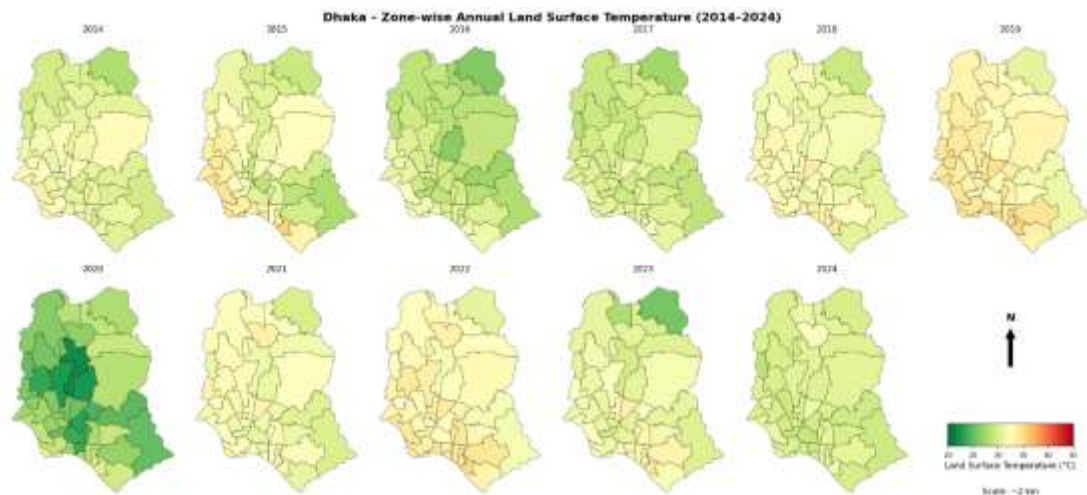


Figure 7.4 a: one-wise Annual LST Maps (2014–2024)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Across the eleven-year period, the maps indicate a gradual upward shift in baseline temperatures, particularly during the post-monsoon and winter months. This suggests a weakening of seasonal cooling and an intensification of background urban heat. The spatial persistence of hotspot zones across multiple years underscores the structural nature of Dhaka’s heat stress, driven by long-term land-use change and urban densification.

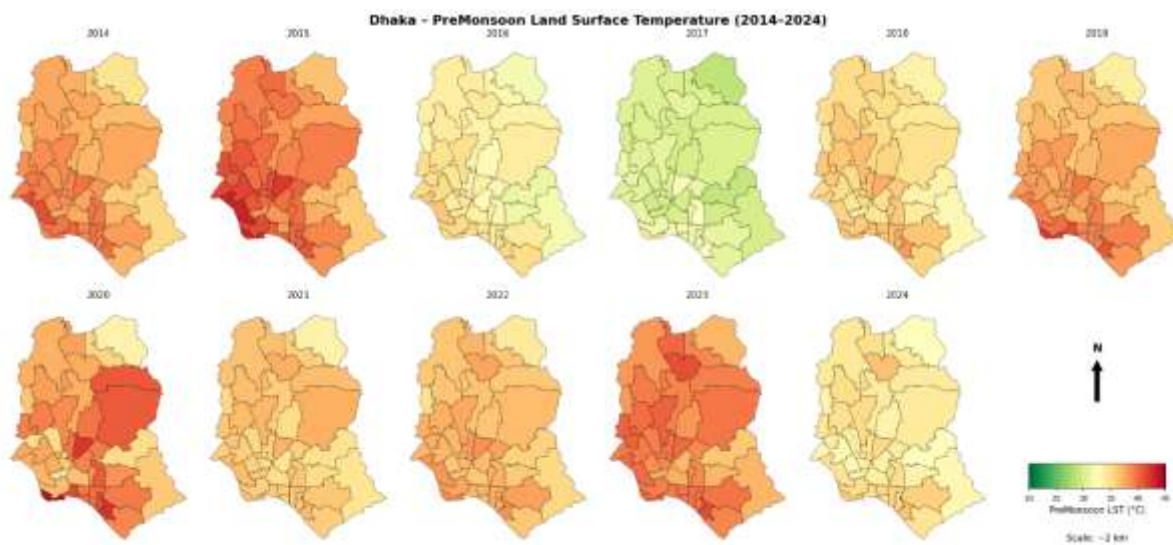


Figure 7.4b: Pre-Monsoon LST Maps (2014–2024)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Pre-Monsoon Heat Intensification

The pre-monsoon LST composites (Figure 7.4 b) highlight the seasonal peak of thermal stress. Over time, more zones transition into higher temperature classes, indicating a spatial expansion of heat-prone areas. The years 2015, 2020, and 2023 stand out as particularly warm, reflecting episodic heat events linked to regional climatic anomalies and rapid urban densification. These patterns reinforce the role of the pre-monsoon season as the critical period for heat exposure in Dhaka.

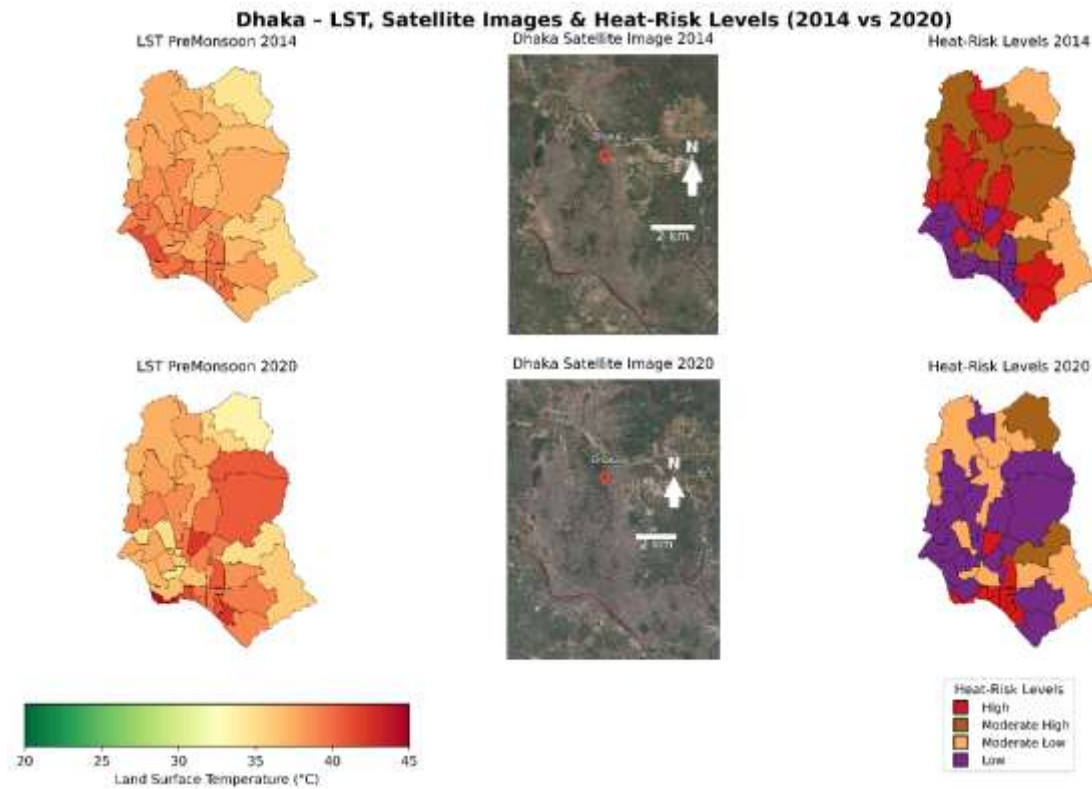


Figure 7.4c: LST, Satellite Imagery & Heat-Risk Comparison (2014 vs 2020)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

A comparative visualization of LST, satellite imagery, and heat-risk levels for 2014 and 2020 (Figure 7.4c) illustrates the spatial expansion of heat-prone areas. The increase in high-risk zones corresponds closely with the densification of built-up surfaces visible in the satellite imagery. This reinforces the link between land-cover transformation and rising thermal exposure.

Point-Based LST–LULC Interaction Analysis

To complement the zone-level and temporal analyses, five representative points (Figure 7.4d) were selected across Dhaka to examine how Land Surface Temperature (LST) responds to variations in land use and land cover (LULC) under different socioeconomic contexts. This point-based approach follows established heat vulnerability frameworks that integrate environmental exposure with social sensitivity (Harlan et al., 2006; Reid et al., 2009; IPCC, 2022). Heat exposure categories were derived from long-term pre-monsoon maximum LST values, hotspot persistence, and trend slopes (Weng, 2009; Li et al., 2013), while poverty classifications were informed by data from the Bangladesh Bureau of Statistics (BBS, 2022), the Dhaka Structure Plan (RAJUK, 2016–2035), and World Bank urban poverty assessments. (Detailed analysis in Appendix 1B).

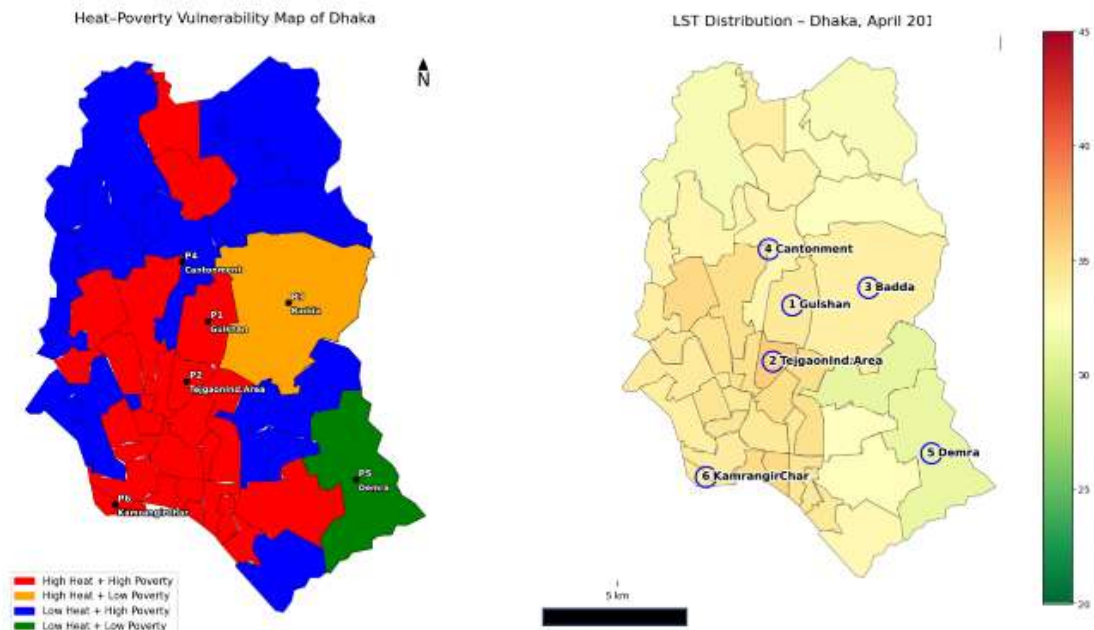


Figure 7.4d: Heat–Poverty Vulnerability Map of Dhaka (left) and LST Distribution for April 2017 (right).

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Vegetation Dynamics and the NDVI–LST Relationship

Vegetation plays a central role in moderating surface temperature through shading and evapotranspiration. Pre-monsoon NDVI maps from 2014–2024 (Figure 7.e) show clear spatial contrasts: peri-urban zones maintain relatively high vegetation density, while central urban zones exhibit consistently low NDVI values. Some zones show gradual vegetation loss over time, particularly in rapidly developing areas.

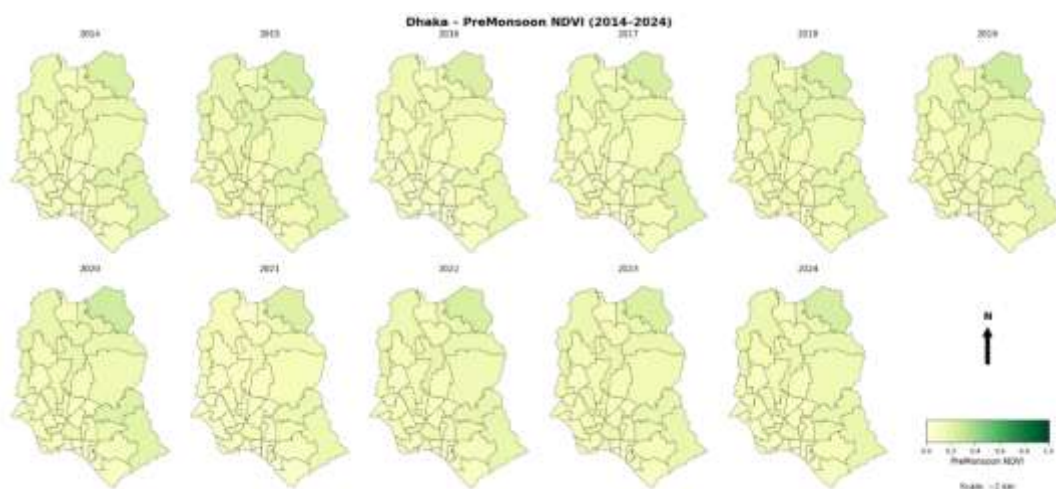


Figure 7.4e: Pre_Monsoon NDVI Maps (2014–2024)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

The NDVI–LST scatter plot (Figure 7.4f) quantifies this relationship, showing a negative correlation ($r = -0.20$) between vegetation density and surface temperature. Although the correlation is modest, it is consistent with the expected cooling effect of vegetation in tropical urban environments. The temporal distribution of points indicates that later year's cluster toward lower NDVI and higher LST values, suggesting a convergence toward more heat-prone conditions.

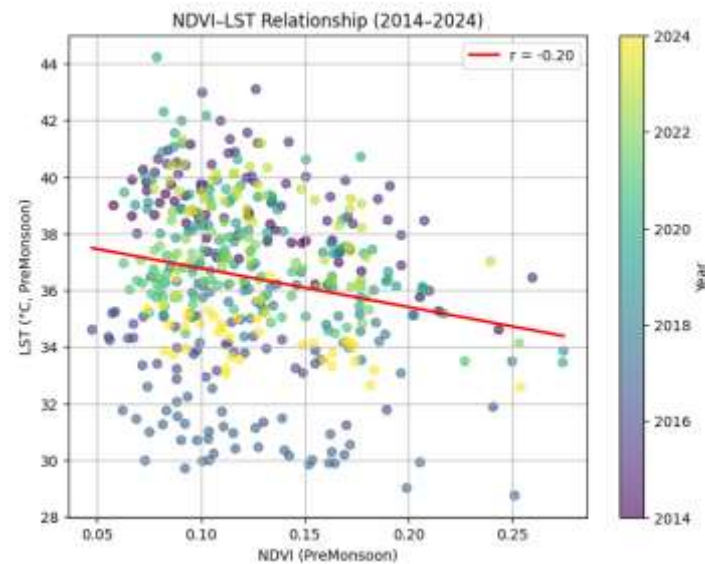


Figure 7.4f: NDVI–LST Scatter Plot (2014–2024)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

The dual-axis trend plot (Figure 7.4g) provides a combined view of NDVI and LST trends over the eleven-year period. NDVI values remain relatively stable, fluctuating within a narrow range, while LST shows a gradual upward trend. This divergence indicates that vegetation cover has not increased sufficiently to counteract rising temperatures. The stability of NDVI alongside increasing LST highlights the limitations of existing green infrastructure in mitigating urban heat.

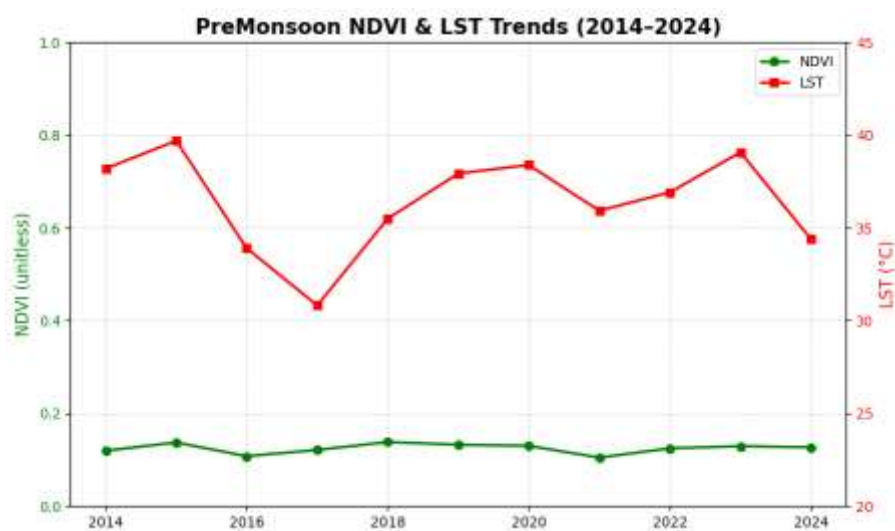


Figure 7.4g: NDVI & LST Dual-Axis Trend Plot (2014–2024)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

The modest but consistent negative NDVI–LST relationship underscores the cooling role of vegetation, yet the stability of NDVI alongside rising LST indicates that existing green cover is insufficient to offset thermal intensification. Together, these findings highlight the need for targeted climate-responsive planning, including strategic greening, reflective materials, and improved ventilation corridors. The results also provide essential context for the vulnerability assessment presented in Section 6.5. Cold Spots (2014–2024).

Extended Analysis (Appendix 1B,1C)

To complement the zone-level and long-term analyses, additional investigations were conducted and are presented in

1. Detailed case study: Gulshan (P1) Multi-temporal imagery and LST trends reveal rapid densification, vegetation loss, and one of the steepest warming rates in the city (+0.51 °C/year).
2. Seasonal heat-risk clustering (K-means) Clustering results confirm that pre-monsoon months exhibit the highest heat risk, while monsoon and winter months show moderated but persistent hotspots.

7.5 Spatial Distribution of Hotspots and Cold Spots

This section analyses the spatial and temporal behaviour of extreme thermal conditions in Dhaka using a percentile-based hotspot–cold spot framework. The assessment is based on 399 Landsat LST images from 2014 to 2024, from which zone-wise mean LST values were extracted. After removing cloudy or invalid scenes, approximately 395 valid zone-year records were used for classification. This approach captures both persistent and episodic thermal extremes and provides a robust basis for identifying heat-sensitive areas.

Thresholds and Classification

Hotspots and cold spots were defined using the 95th and 5th percentiles of the full LST distribution, corresponding to ≥ 40.38 °C and ≤ 21.71 °C respectively. Each zone-image record was assigned to one of three categories: hotspot, cold spot, or normal. Across the decade, 20 hotspot events and 20 cold-spot events were identified, with the remaining records falling within the normal range. These values

represent thermal events rather than unique zones, meaning that a single zone may appear as a hotspot or cold spot in multiple years.

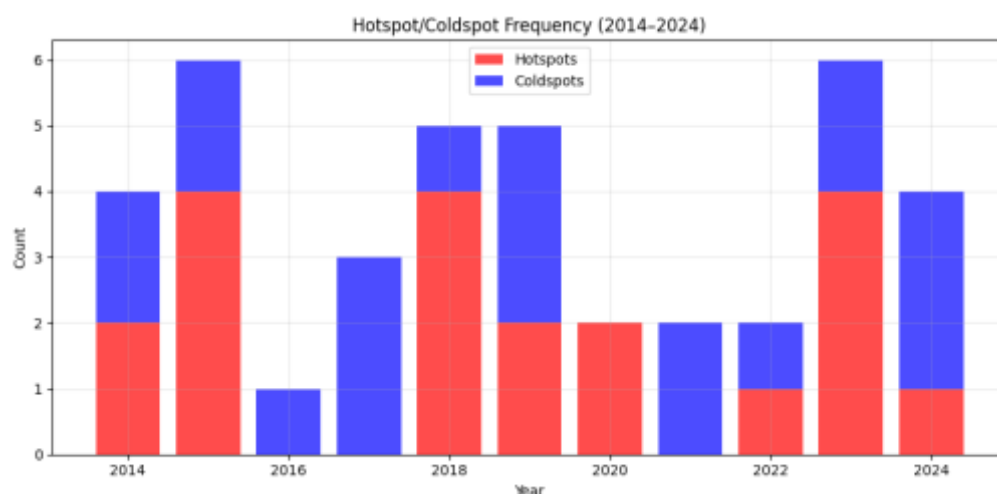


Figure7.5a. Annual frequency of hotspot and cold-spot events (2014–2024)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Hotspots and Cooling Deficit Index (CDI)

Spatial patterns derived from percentile classification and the Cooling Deficit Index (CDI) reveal a clear thermal hierarchy across Dhaka. Hotspots are consistently concentrated in dense urban and industrial zones such as Mirpur, Tejgaon, Rampura, Jatrahari, and Mohammadpur. These areas are characterised by high building density, extensive impervious surfaces, limited vegetation, and substantial anthropogenic heat emissions. Hotspot areas expand noticeably over the decade, reflecting intensifying urban heat island effects.

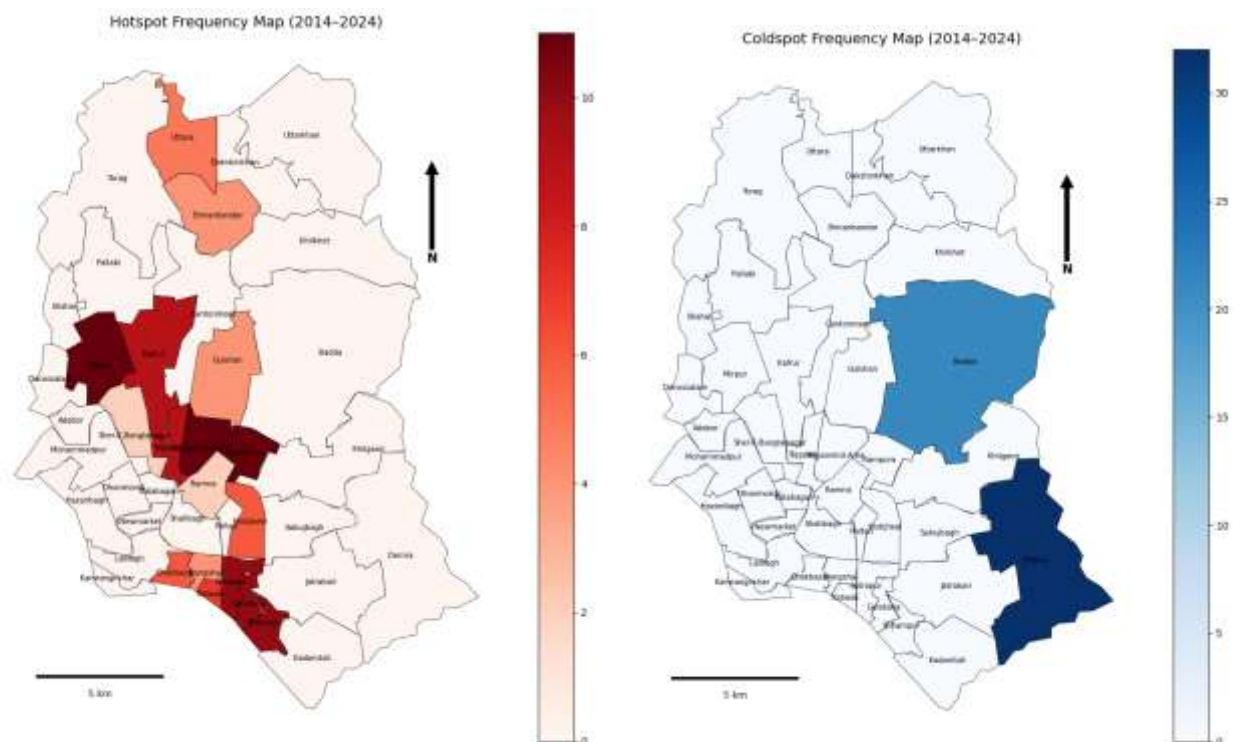


Figure 7.5b. Structural Patterns of Hotspot Persistence and Cooling Deficits in Dhaka (2014–2024)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Cold spots, in contrast, are primarily located around Hatirjheel, Ramna Park, the Botanical Garden, and the Buriganga and Balu River corridors. These areas benefit from vegetation, open water, and permeable surfaces, which support local cooling. However, cold-spot zones show gradual shrinkage, consistent with ongoing land conversion and vegetation loss.

CDI-Based Spatial Contrast

The Cooling Deficit Index (CDI), defined as normalized LST minus normalized NDVI, provides an integrated measure of thermal stress. High CDI values align closely with hotspot zones, while low CDI values correspond to cold-spot areas. CDI intensifies over time, particularly in central and industrial zones, confirming that thermal stress is structurally embedded in Dhaka's urban form. The full annual CDI series is provided in Appendix 1D.

Persistence and Transition of Hotspots

Hotspot persistence and transition diagnostics (Figure 7.5c) show that Tejgaon, Mirpur, Rampura, and Jatrahari function as long-term hotspots, while transitional hotspots emerge in Badda, Khilgaon, and Demra. Cold-spot stability is strongest in Hatirjheel, Ramna Park, and the Botanical Garden. CDI trends indicate increasing thermal stress in zones undergoing rapid densification.

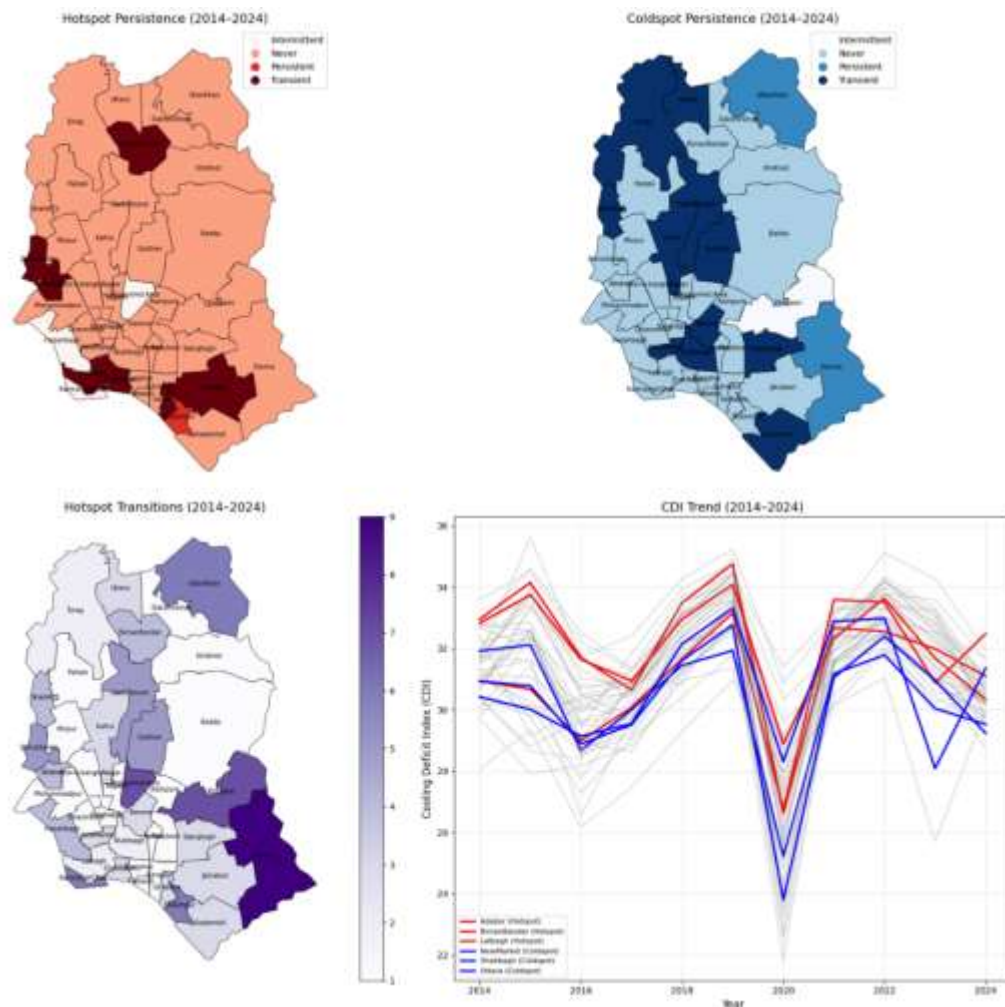


Figure 7.5c Structural patterns of hotspot persistence and cooling deficits in Dhaka (2014–2024)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Overall, the hotspot–cold-spot analysis demonstrates that thermal extremes in Dhaka are neither random nor short-lived. Hotspots are expanding and intensifying in dense urban cores, cold spots are shrinking as green and water-rich areas come under pressure, and spatial inequality in heat exposure is increasing.

These patterns provide essential context for the subsequent assessment of heat–poverty interactions. These findings provide essential context for the heat–poverty interaction analysis presented in the next chapters.

7.6 Spatial Distribution of Poverty (2017)

The spatial distribution of poverty in Dhaka exhibits a clear and persistent socioeconomic gradient shaped by rapid urbanisation, high population density, and uneven access to services. Using NB_BUDENS (building density) as a proxy for socioeconomic vulnerability, the 2017 poverty map (Figure 7.6a) identifies wards characterised by compact built-up structures, limited open space, and extensive informal housing. High-vulnerability areas are concentrated in the older and densely populated parts of the city, including Lalbagh, Kotwali, KamrangirChar, Hazaribagh, Badda, Rampura, Mirpur, Mohammadpur, and Jatrabari. These wards exhibit the highest building density and represent long-standing pockets of deprivation.

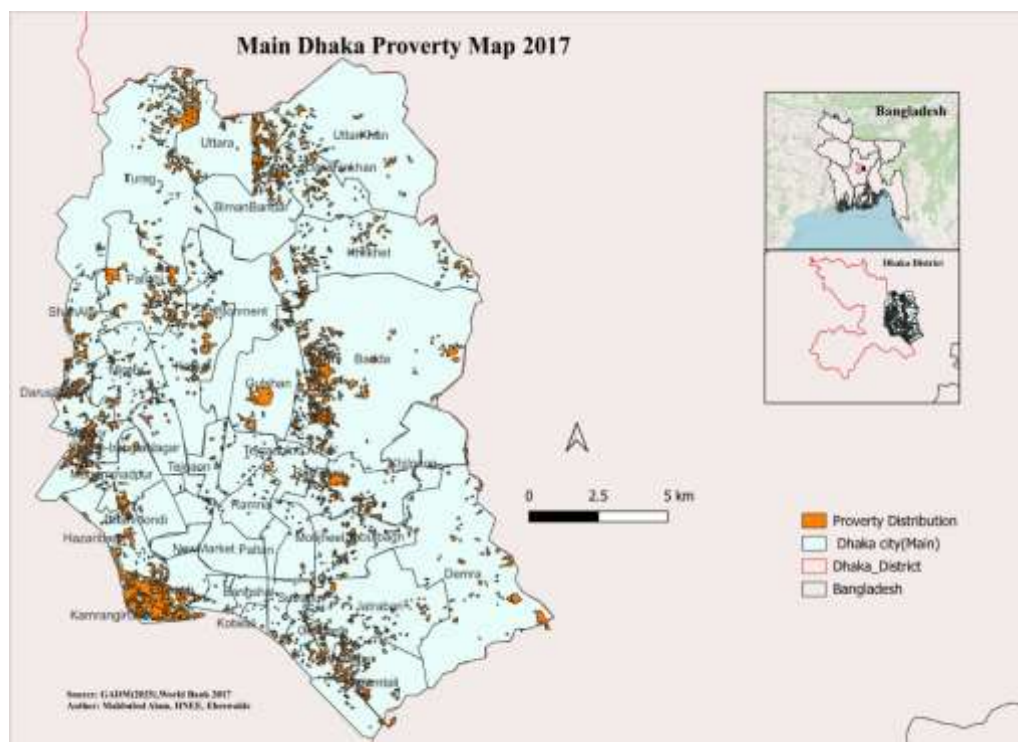


Figure 7.6a: Spatial Distribution of Poverty 2017

In contrast, planned neighbourhoods such as Gulshan, Banani, Baridhara, Dhanmondi, Uttara, and the Cantonment display substantially lower NB_BUDENS values. These areas benefit from wider roads, improved infrastructure, and greater access to services, resulting in lower socioeconomic vulnerability.

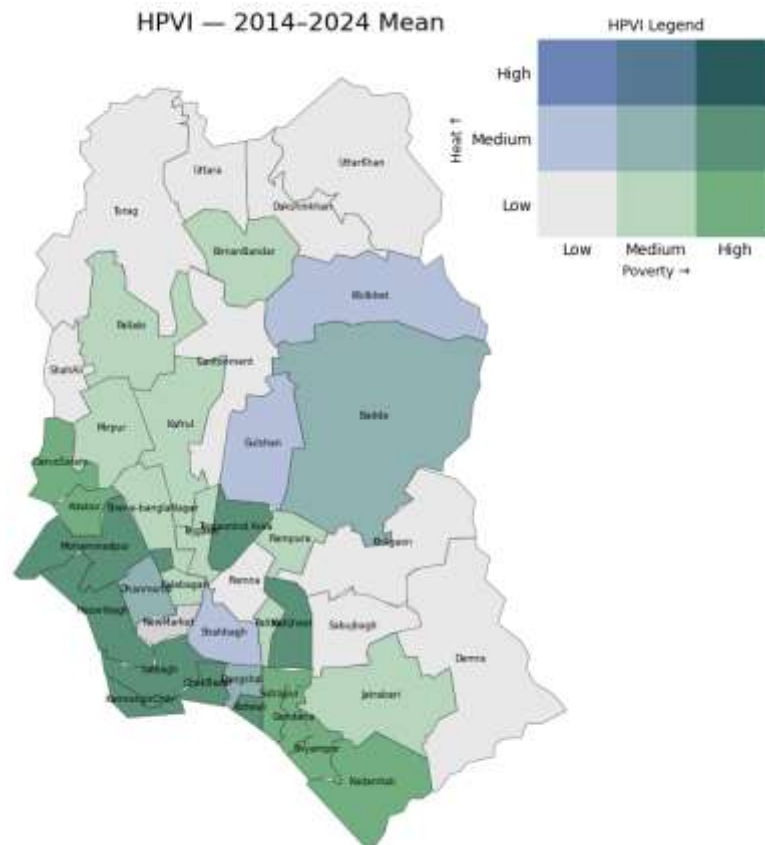


Figure 7.6b : HPVI (Building Density- NB_BUDENS)

Source: GEE and Python processing. Map by Mahbubul Alam (2026).

Figure 6.6a also serves as the administrative reference map for Dhaka, providing ward labels that support geographic orientation throughout this section. This contextual map establishes the spatial framework for interpreting the subsequent analytical panels. Using building density as a proxy for socioeconomic vulnerability reveals three consistent patterns:

1. High building density in older parts of Dhaka (Lalbagh, Kotwali, KamrangirChar)
2. Dense informal settlements in Mirpur, Mohammadpur, Rampura, and Jatrabari
3. Lower density and lower vulnerability in Gulshan, Banani, Baridhara, and Uttara

This spatial pattern aligns with known socioeconomic disparities, confirming NB_BUDENS as a reliable proxy for vulnerability.

The poverty distribution pattern shown in Figure 6.6b indicates that several wards exhibit markedly higher NB_BUDENS values, reflecting concentrated socioeconomic vulnerability. The most affected areas include Badda, KamrangirChar, Hazaribagh, and Jatrabari, all characterised by dense built-up structures, limited open space, and constrained access to essential services. These wards represent long-standing pockets of deprivation where informal housing, inadequate infrastructure, and environmental stressors converge.

In contrast, central and northern wards such as Gulshan, Cantonment, and Dhanmondi display substantially lower NB_BUDENS values. These areas benefit from planned urban development, better infrastructure, and greater access to services, resulting in improved living conditions and reduced vulnerability.

Panels (b) and (c) of Figure 6.6c complement the poverty analysis by illustrating the spatial distribution of mean land surface temperature (LST) and the overlay of slum boundaries on LST.

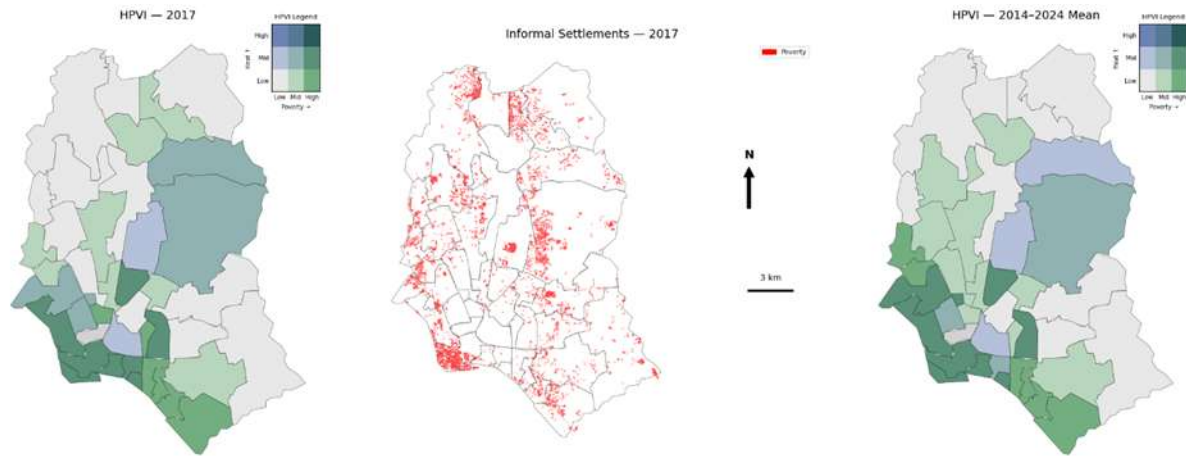


Figure 7.6c Spatial Distribution and Persistence of Socioeconomic Vulnerability in Dhaka

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Although these panels omit ward labels to maintain visual clarity, they reveal a consistent pattern: many high-poverty wards coincide with elevated thermal exposure and dense informal settlements. This spatial overlap underscores the compounded vulnerability faced by low-income communities, where socioeconomic disadvantage and environmental stress reinforce one another.

Overall, the 2017 poverty distribution provides a critical baseline for understanding the broader vulnerability landscape of Dhaka. The spatial gradient observed here—ranging from high-density, underserved southern and western wards to more affluent, planned northern zones—forms the foundation for interpreting heat exposure, slum concentration, and environmental risk in the subsequent sections of this chapter.

7.7 Heat–Poverty Interaction

The interaction between heat exposure and socioeconomic vulnerability is examined using ward-level data for 2017. This analysis integrates environmental indicators (LST, NDVI, NDBI, MNDWI) with a poverty proxy based on building density (NB_BUDENS). Together, these variables reveal how environmental stress and socioeconomic disadvantage overlap across Dhaka’s urban landscape. To maintain clarity in the main chapter, extended scatter plots, supplementary correlation panels, and high-resolution diagnostic figures supporting this section are provided in Appendix.

Correlation Analysis Between LST and Environmental Indicators

This correlation analysis is based on 451 (Appendix 2) random data points, representing ward-level observations across Dhaka. To understand the environmental drivers of surface heat in Dhaka, correlation analysis was conducted between Land Surface Temperature (LST) and four key variables: NDVI (vegetation), NDBI (built-up intensity), MNDWI (water presence), and IBI (impervious surface index). All variables were analysed using 2017 data to ensure spatial consistency with the poverty baseline.

Figure 7.7a presents the correlation matrix heatmap, summarizing the pairwise relationships among the selected variables. The strongest correlation is observed between LST and MNDWI ($r = -0.69$), indicating that water presence is associated with lower surface temperatures. NDVI also shows a weak negative correlation with LST ($r = -0.12$), while NDBI and IBI show negligible correlation values ($r \approx 0.00$), suggesting limited explanatory power in this dataset. The matrix also reveals a moderate negative correlation between MNDWI and NDVI ($r = -0.52$), reflecting the spatial separation between vegetated and water-dominated zones.

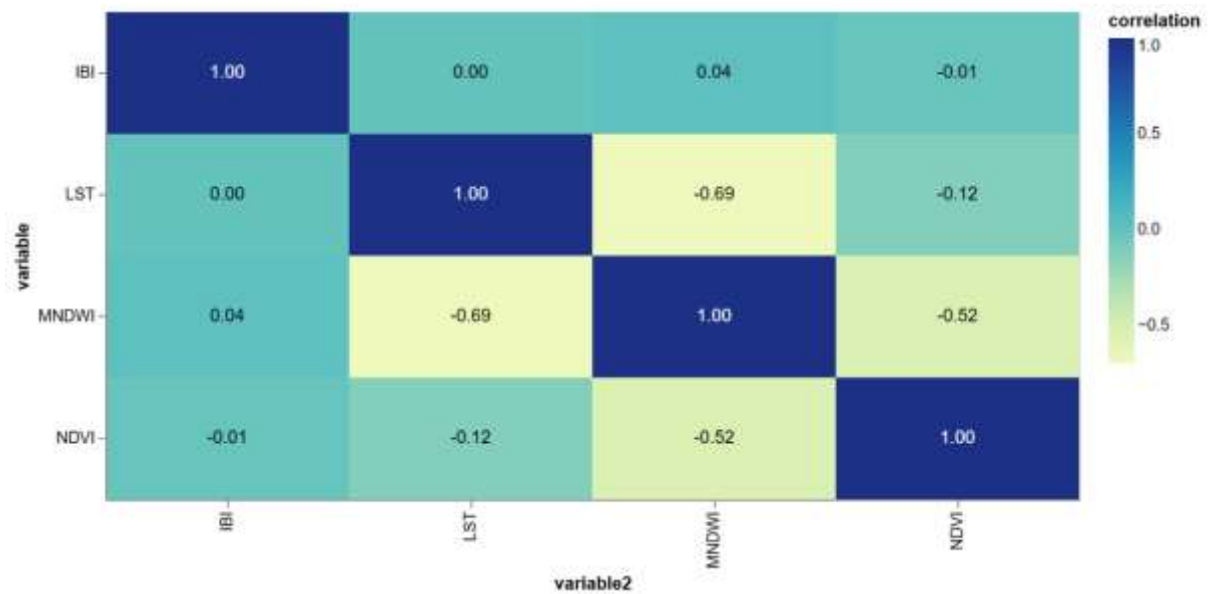


Figure 7.7a: Correlation Matrix Heatmap

LST–NDVI: Weak Negative Correlation

Figure 7.7b displays the scatter plot between Normalized Difference Vegetation Index (NDVI) and Land Surface Temperature (LST) for areas where $LST \geq 20^{\circ}\text{C}$. The regression line has a slope of 7.393 and $R^2 = 0.004$, indicating a very weak positive relationship. While vegetated zones such as Ramna Park and Hatirjheel exhibit relatively lower LST values, the overall trend suggests that NDVI has minimal influence on surface temperature across the study area. This limited cooling effect is likely due to the low density and fragmented distribution of vegetation throughout Dhaka, which reduces the potential for widespread thermal regulation.

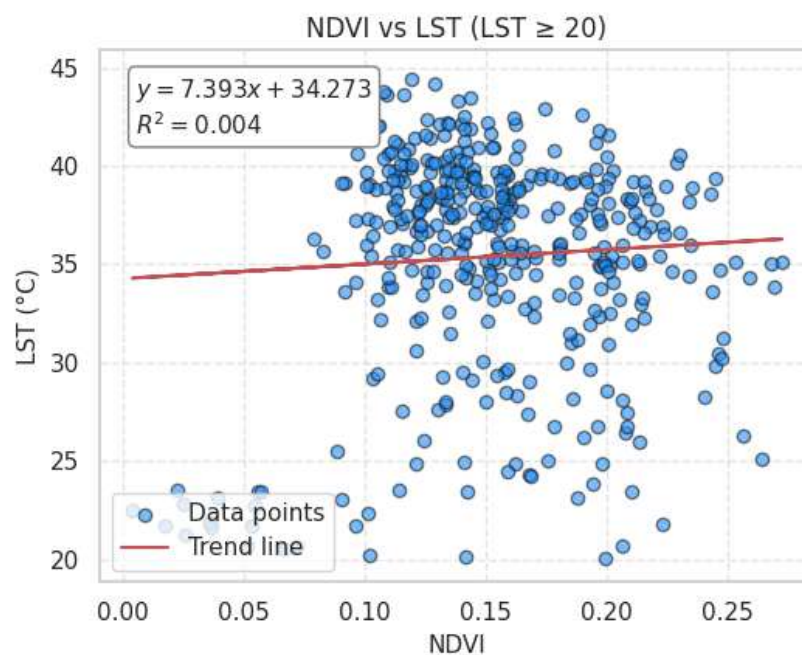


Figure 7.7b: NDVI vs LST ($LST \geq 20^{\circ}\text{C}$)

LST–NDBI: Moderate Positive Correlation

Figure 7.7c presents the scatter plot between NDBI and LST ($LST \geq 20^\circ\text{C}$). The regression slope is +89.55 with $R^2 = 0.343$, indicating a moderate positive correlation. Built-up zones such as Tejgaon, Mirpur, and Kamrangirchar consistently show elevated LST values, confirming that impervious surfaces contribute significantly to urban heat.

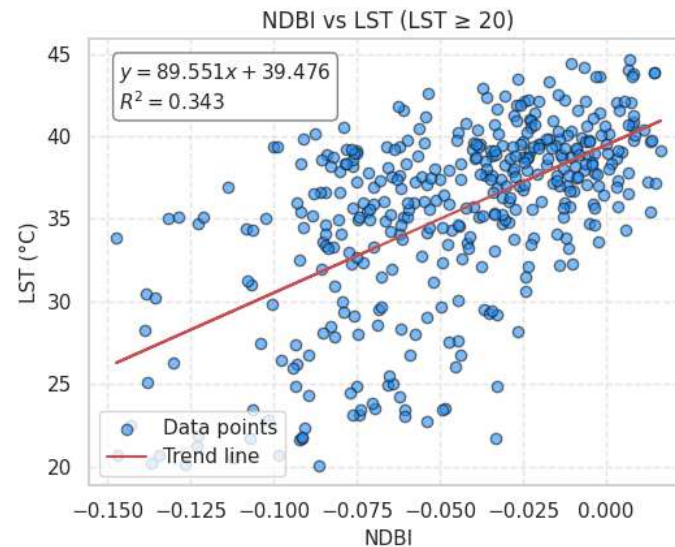


Figure 7.7c: NDBI vs LST ($LST \geq 20^\circ\text{C}$)

LST–MNDWI: Moderate Negative Correlation

Figure 7.7d illustrates the scatter plot between MNDWI and LST ($LST \geq 20^\circ\text{C}$). The regression slope is -87.16 with $R^2 = 0.421$, indicating a moderate negative correlation. Areas with higher water presence consistently show lower LST values, highlighting the strong cooling influence of water bodies, although this effect is spatially constrained due to ongoing land conversion.

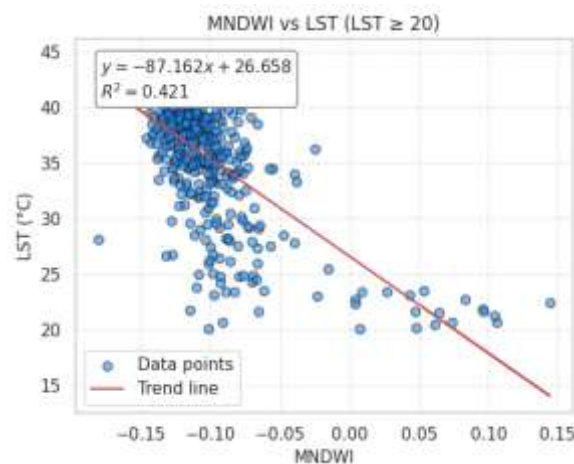


Figure 7.7d: LST–MNDWI: Moderate Negative Correlation

LST–Poverty Index (Building Density): No Correlation

Figure 7.7e presents the scatter plot between the poverty index (proxied by building density) and LST ($LST \geq 20^\circ\text{C}$). The regression slope is $+0.115$ with $R^2 = 0.000$, indicating no measurable correlation. Despite expectations that densely built, low-income areas might exhibit elevated surface temperatures, the data show no consistent relationship. This suggests that building density alone is insufficient to explain LST variations and that other factors—such as vegetation cover, ventilation, and material reflectivity—may play a more decisive role in shaping urban heat exposure.

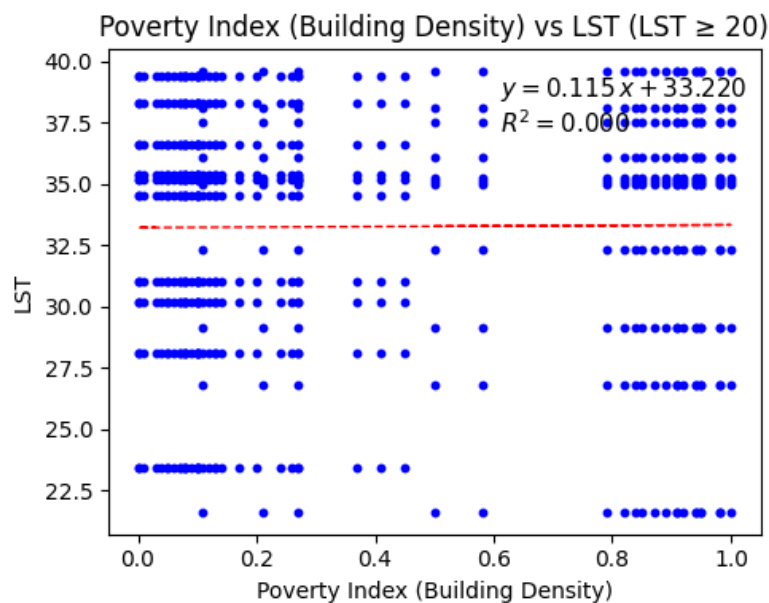


Figure 7.7e: LST–Poverty Index (Building Density)

This chapter shows that Dhaka’s surface heat is driven mainly by environmental conditions rather than poverty or building density alone. Water bodies and vegetation provide clear cooling effects, while impervious built-up areas intensify heat. In contrast, poverty indicators based on building density show no meaningful relationship with LST. These results highlight that effective heat-mitigation strategies must prioritize green–blue infrastructure and improved urban design to reduce heat exposure in vulnerable areas.

Comparison of Poor vs Non-Poor Areas

To assess thermal inequality across Dhaka’s urban landscape, this section compares heat and environmental indicators between wards classified as poor (high building density) and non-poor (low building density). The analysis draws on multiple figures that examine mean land surface temperature (LST), hotspot frequency, vegetation cover (NDVI), water presence (MNDWI), and long-term temperature trends.

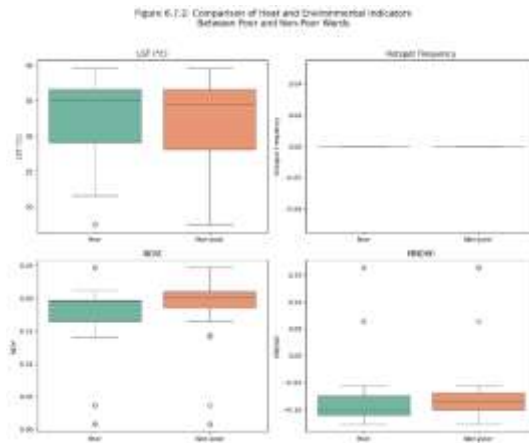


Figure 7.7 f: Comparison of Poor vs Non-Poor Areas

Figure 7.7f presents box plots comparing poor and non-poor wards across four key indicators. Poorer wards consistently exhibit higher mean LST values, often 1–2°C warmer than their non-poor counterparts. Vegetation cover is substantially lower in poor areas, as reflected by lower NDVI values, while water bodies—captured by MNDWI—are scarce or absent across both groups. Hotspot frequency shows no measurable difference, suggesting that while extreme heat events are spatially distributed, chronic heat exposure is concentrated in vulnerable zones.

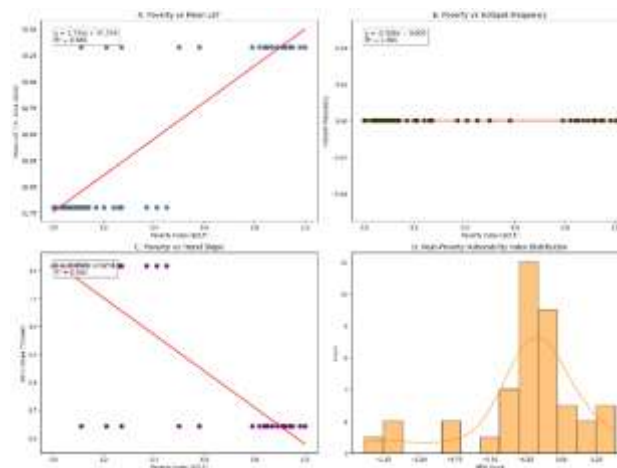


Figure 7.7g : Multi-panel poverty correlations plots

Supporting this, the multi-panel figure “Poverty vs Heat Indicators” reveals a strong positive correlation between poverty index and mean LST ($R^2 = 0.656$), indicating that poorer communities endure higher

baseline temperatures. Interestingly, the trend slope analysis shows a negative correlation ($R^2 = 0.656$), suggesting that poor areas—already thermally saturated—exhibit slower warming over time, likely due to limited land-cover change. The Heat–Poverty Vulnerability Index (HPVI) distribution further confirms that most vulnerable scores cluster in poorer wards, reinforcing the spatial concentration of heat risk

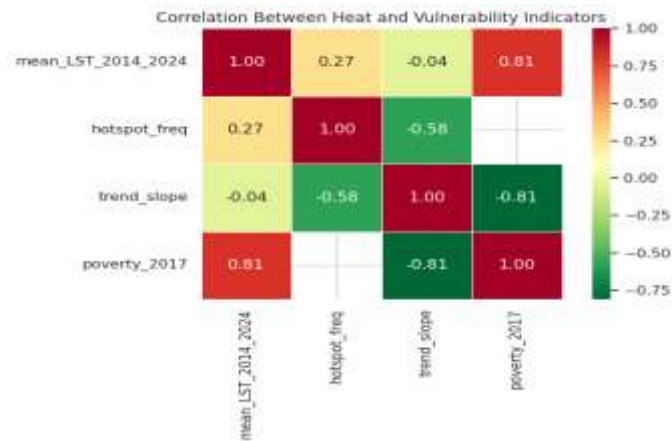


Figure 7.7h: Poverty vs Heat correlation heatmap

A correlation heatmap consolidates these relationships, showing a strong positive correlation between poverty and mean LST ($r = 0.81$), and a strong negative correlation between poverty and LST trend slope ($r = -0.81$). These findings highlight that while poor areas are not necessarily warming faster, they are already significantly hotter and more environmentally degraded.

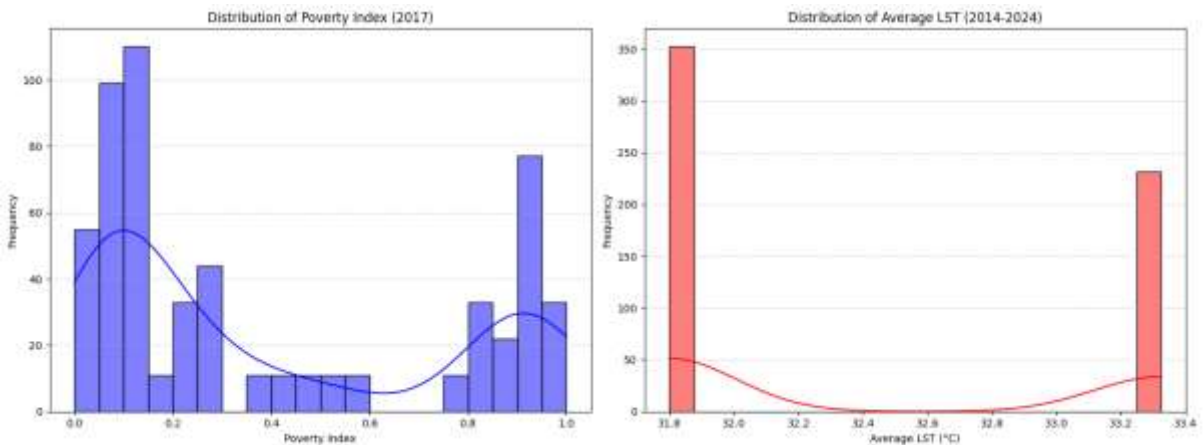


Figure 7.7i: Distribution of Poverty Index and Average LST

Finally, Figure 7.7i Distribution of Poverty Index and Average LST illustrates the underlying data distributions. The poverty index is right-skewed, with most wards falling into low-income categories, while the LST distribution is bimodal, indicating distinct thermal zones across the city. This supports the conclusion that heat exposure is not evenly distributed and disproportionately affects poorer communities.

Together, these figures demonstrate a clear pattern of thermal inequality: poorer communities in Dhaka face disproportionately higher heat stress, compounded by reduced vegetation, minimal water access, and limited adaptive capacity. This underscores the urgent need for targeted climate resilience strategies in vulnerable urban zones.

Identification of High-Risk Zones

The HPVI quantile map (Figure 6.7) identifies clear spatial clusters of heat–poverty vulnerability across Dhaka. Wards classified as *Very High* and *High* vulnerability represent locations where elevated temperatures, limited green–blue infrastructure, and socioeconomic disadvantage intersect, resulting in compounded exposure risks.

Very High vulnerability zones are concentrated in long-established dense settlements and industrial belts. Several wards in Mirpur exhibit persistently high LST and compact residential blocks. The Tejgaon Industrial Area shows extensive impervious surfaces and minimal vegetation, contributing to sustained heat retention. Neighbourhoods such as Rampura, Badda, Khilgaon, and Shyampur are characterised by dense built-up structures, poor ventilation, and limited open space. Central riverside wards including KamrangirChar, Kotwali, Sutrapur, and Gendaria display high levels of informal housing and low cooling capacity, further intensifying vulnerability.

High vulnerability zones include rapidly urbanising northern wards such as Uttara and Dakshinkhan, where expanding built-up cover contributes to rising temperatures. Mixed-use and commercial areas such as BimanBandar and Tejgaon retain heat due to extensive impervious surfaces. Dense commercial districts—Ramna, Shahbagh, New Market, Paltan, and Motijheel—experience high daytime heat exposure. Historic core areas such as ChawkBazar and Bangshal, with narrow streets and high population density, also fall within this category.

Moderate vulnerability wards, including Cantonment, Gulshan, Lalbagh, Hazaribagh, and Adabor, benefit from planned layouts or mixed land-use patterns that provide partial resilience. Low vulnerability areas—such as Mohammadpur, Dhanmondi, Kalabagan, Turag, Pallabi, Sabujbagh, Demra, Kadamtali, and Jatrabari—exhibit better vegetation cover, lower poverty levels, or improved infrastructure, reducing overall exposure.

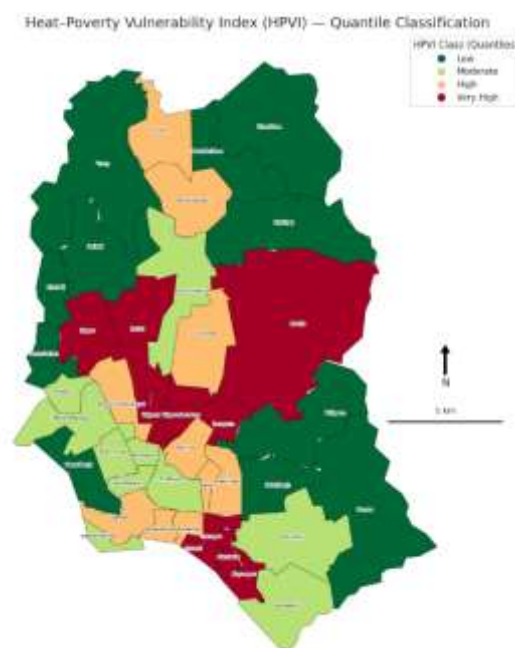


Figure 7.7j: HPVI quantile map

Source: GEE and Python processing. Map by Mahbubul Alam (2026).

Supplementary Statistical Evidence of Thermal Inequality

A comparison between the highest- and lowest-poverty quartiles shows that high-poverty wards experience higher mean surface temperatures (30.30 °C) than low-poverty wards (29.32 °C), a difference of 0.99 °C. Despite its modest size, this gap is statistically significant, as confirmed by both the independent t-test ($t = 4.25$, $p = 0.00050$) and the Mann–Whitney U test ($U = 91$, $p = 0.0022$). The accompanying multi-panel figure reinforces these findings: the scatterplot and regression line show a clear positive relationship between poverty and LST, the correlation matrix reports a strong association ($r \approx 0.73$), and the boxplot illustrates consistently higher temperatures in high-poverty wards. Together, these results provide concise and robust evidence of systematic thermal inequality across Dhaka.

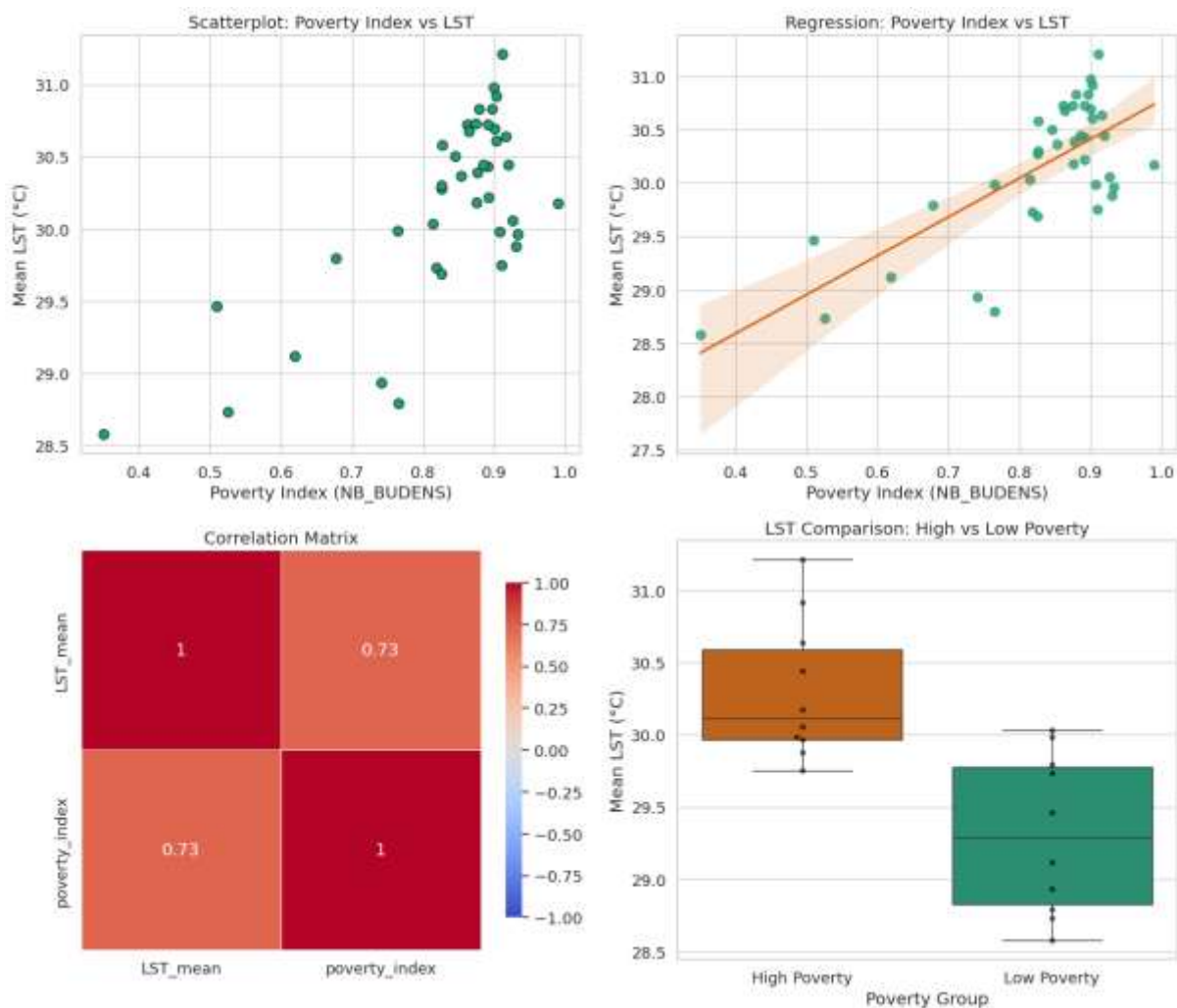


Figure7.7 k: Statistical Evidence of Thermal Inequality

Source: GEE and Python processing. plot by Mahbubul Alam (2026).

Overall, the spatial pattern demonstrates that the highest-risk zones are those where heat stress and socioeconomic disadvantage overlap, underscoring the structural nature of thermal inequality in Dhaka.

8. Discussion

The results presented in Chapter 6 demonstrate that Dhaka's thermal environment is shaped by a combination of seasonal climatic patterns, urban morphological characteristics, and socioeconomic disparities. The analysis reveals a pronounced pre-monsoon thermal peak that recurs consistently throughout the study period and is clearly visible in the monthly LST patterns (**see Figure 7.1b**). High-density and low-income wards exhibit persistently elevated surface temperatures, whereas areas with substantial vegetation or proximity to water bodies remain comparatively cooler. Although the temperature difference between the highest- and lowest-poverty quartiles is less than one degree Celsius, it is statistically significant and highly relevant in the context of Dhaka's dense built environment. The long-term persistence of hotspots, particularly in industrial and compact residential zones (**Figure 7.5b**), indicates that thermal stress is structurally embedded rather than driven by short-term climatic fluctuations.

These findings are consistent with previous studies documenting the relationship between rapid urbanisation, vegetation loss, and surface heating in Dhaka (Imran et al., 2021; Faisal et al., 2022). The strong correlation between impervious surfaces and LST aligns with international evidence on urban heat island dynamics (Trotter et al., 2017; Mathew et al., 2016). Likewise, the cooling influence of green and blue infrastructure corresponds with global research demonstrating the moderating effects of vegetation and water bodies (Bowler et al., 2010; Hassan et al., 2021). The observed heat–poverty relationship reinforces local findings on thermal stress in informal settlements (Hossain et al., 2022) and reflects broader literature on the vulnerability of low-income populations to heat exposure (Nahar et al., 2010; Kjellstrom et al., 2009).

The mechanisms underlying these patterns are clear. High building density and extensive surface sealing increase heat storage and reduce nocturnal cooling. Limited vegetation diminishes shading and evapotranspiration, while narrow street canyons restrict airflow and intensify heat retention. These processes are evident in the seasonal LST variations (Figure 6.2c) and in the spatial distribution of cold spots near rivers and wetlands (Figure 6.4c). Together, they illustrate that thermal inequality in Dhaka arises from both physical land-surface processes and socio-spatial structures.

The implications for urban planning are substantial. Measures to reduce heat exposure must prioritise neighbourhoods where thermal stress and socioeconomic vulnerability coincide. The protection and restoration of green and blue spaces, the improvement of urban ventilation corridors, and the integration of heat-risk mapping into planning processes are essential. The findings highlight that urban heat in Dhaka is not only an environmental challenge but also a question of social equity.

9. Limitations and Future Work

9.1 Limitations

Although this study provides a comprehensive assessment of Dhaka's thermal environment over a ten-year period, several methodological and data-related limitations must be acknowledged. These limitations do not undermine the overall findings but help contextualise the results.

First, the analysis relies primarily on Landsat 8/9 thermal infrared data, which are affected by cloud contamination and a 16-day revisit cycle. These constraints reduce temporal coverage, particularly during the monsoon season, and may influence seasonal or annual composites. LST retrieval also depends on atmospheric correction and emissivity assumptions, which introduce uncertainties at the pixel level.

Second, the 30 m spatial resolution of Landsat is insufficient to capture micro-scale thermal variability within dense neighbourhoods or informal settlements. Fine-scale heterogeneity—such as rooftop materials, shading patterns, and courtyard vegetation—remains underrepresented. Higher-resolution thermal datasets (e.g., ECOSTRESS, ASTER, UAV-based sensors) could provide more detailed insights.

Third, the Mann–Kendall and Theil–Sen trend analyses are sensitive to missing data and interannual variability. Years with limited cloud-free observations may obscure subtle warming or cooling signals, and non-significant trends should therefore be interpreted with caution.

Fourth, the socioeconomic proxy used in this study (NB_BUDENS) captures only one dimension of vulnerability. While building density correlates with overcrowding and poverty in Dhaka, it does not reflect multidimensional factors such as income, age structure, health conditions, access to cooling infrastructure, or occupational exposure.

Fifth, the temporal mismatch between the LST dataset (2014–2024) and the socioeconomic dataset (2017) introduces uncertainty, even though building density typically changes slowly.

Finally, the study does not incorporate air temperature, humidity, wind speed, or anthropogenic heat emissions. These variables are important for assessing human thermal comfort and may influence the spatial distribution of heat exposure.

9.2 Future Work

Several research directions could build on the findings of this study. Multi-sensor validation using MODIS, Sentinel-3 SLSTR, or ECOSTRESS would strengthen confidence in the observed trends and help distinguish climatic signals from sensor-specific effects. Additional remote-sensing indicators—such as albedo, impervious surface fraction, and soil moisture—could improve understanding of the drivers of persistent hotspots.

The heat–poverty relationship could be refined by integrating openly available socioeconomic datasets such as WorldPop, SEDAC deprivation indices, or GHSL layers. Scenario-based modelling would allow assessment of how different urban-growth trajectories—such as densification, greening, or continued sprawl—may influence future thermal conditions.

Comparative studies with other tropical megacities (e.g., Kolkata, Manila, Jakarta) could situate Dhaka within broader regional patterns of thermal inequality.

10. Conclusion

This report demonstrates that Dhaka's thermal environment is characterised by persistent seasonal heating, expanding structural hotspots, and a clear association between surface temperature and indicators of socioeconomic vulnerability. Because direct income data at ward level are not available for Dhaka, building density (NB_BUDENS) was used as a proxy for socioeconomic vulnerability. This approach is consistent with previous studies and reflects the strong association between overcrowding, informal housing, and poverty in the city. The spatial patterns of LST align closely with this proxy, reinforcing its validity and supporting the interpretation that structurally disadvantaged wards experience disproportionately higher thermal stress.

High-density wards consistently exhibit elevated LST values, while vegetated and water-rich areas provide measurable cooling benefits. The long-term analysis confirms that thermal exposure is unevenly distributed and closely linked to patterns of urban development and social vulnerability. Industrial zones, compact residential areas, and rapidly urbanising peri-urban regions show persistent or intensifying hotspots, whereas cooling effects remain concentrated around rivers, wetlands, and major parks.

Although the study is subject to limitations related to data availability, spatial resolution, and the use of a single socioeconomic proxy, the overall results are robust. The findings underscore the need for climate-equitable urban planning that prioritises structurally vulnerable neighbourhoods, protects remaining green and blue spaces, and integrates heat-risk mapping into municipal decision-making. Measures such as targeted greening, improved ventilation corridors, and nature-based cooling solutions are essential for reducing thermal stress in the most affected areas.

By providing a spatially explicit and methodologically transparent assessment of Dhaka's thermal landscape, this report contributes to a deeper understanding of how environmental stress and socioeconomic vulnerability intersect in rapidly growing tropical cities. The findings offer a foundation for targeted interventions and support the development of evidence-based strategies to enhance climate resilience.

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Appendix

Appendix 1

1A. Seasonal Heat-Risk Classification (K-Means, 2017)

Heat-Risk Classification

Figure 16 shows monthly heat-risk classification using K-means clustering. Zones are categorized into four risk levels: Low, Lower-Medium, Medium, and High. High-risk zones persist across pre-monsoon months, aligning with hotspot zones identified in LST maps. This classification supports targeted urban planning and climate adaptation strategies.

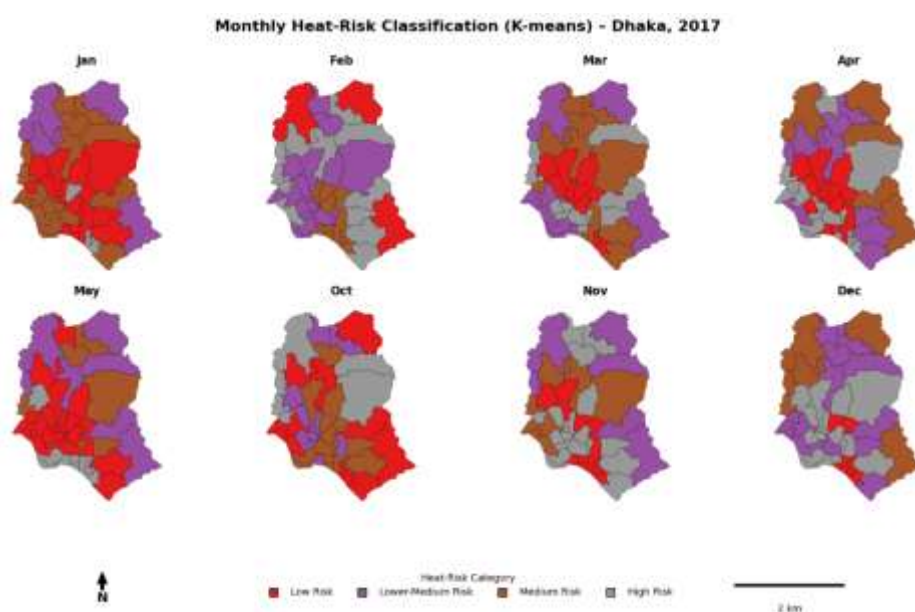


Figure 1a: Monthly Heat-Risk Classification (K-means)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

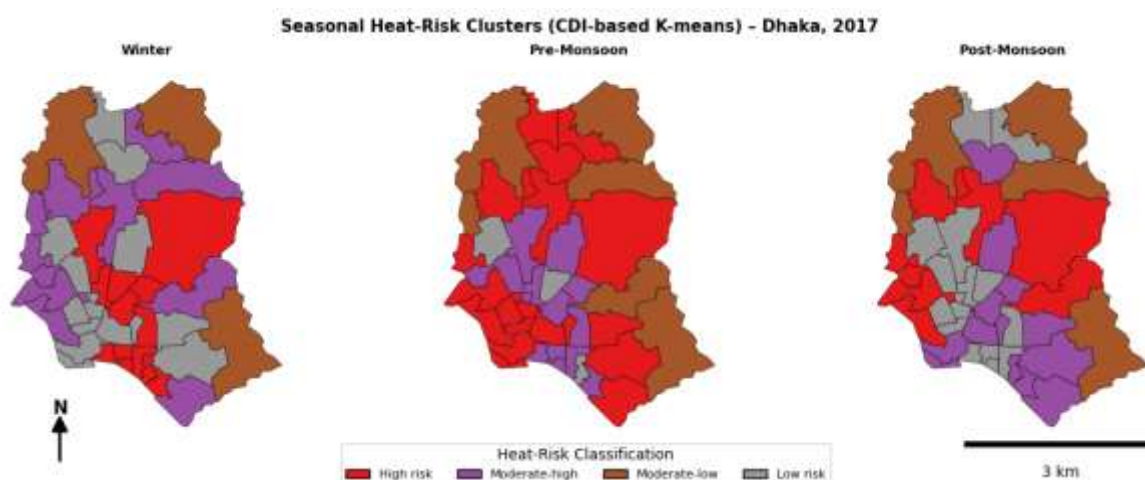


Figure 1b: Monthly Heat-Risk Classification (K-means)

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Interpretation

The spatio-temporal analysis reveals persistent hotspot zones in dense urban and industrial areas, while rural and vegetated zones consistently act as cool spots. Seasonal transitions are sharpest between March–April and November–December, reflecting Dhaka’s monsoon-driven climate. The application of a 5 °C threshold clarifies the spatial hierarchy of thermal stress: only a limited number of zones exceed this threshold, and these correspond to areas with high imperviousness, low vegetation, and strong anthropogenic heat emissions. In contrast, vegetated zones rarely.

1B. Point-Based LST–LULC Interaction Analysis

To complement the zone-level and temporal analyses, five representative points were selected across Dhaka to examine how Land Surface Temperature (LST) responds to different land-use and land-cover (LULC) conditions and socioeconomic contexts. This approach follows established heat-vulnerability frameworks that integrate environmental exposure with social sensitivity (Harlan et al., 2006; Reid et al., 2009; IPCC, 2022). Heat categories were derived from long-term pre-monsoon maximum LST, hotspot persistence, and trend slopes (Weng, 2009; Li et al., 2013), while poverty categories were informed by BBS (2022), the Dhaka Structure Plan (RAJUK, 2016–2035), and World Bank urban poverty assessments. . Five representative points were selected across Dhaka to examine how Land Surface Temperature (LST) responds to different land-use and land-cover (LULC) conditions and socioeconomic contexts based on LST distribution, April 2017

Table 1 . Classification of Selected Points Based on Heat Exposure and Poverty Context

Point	Zone	Category
P1	Gulshan	High Heat + High Poverty
P2	Tejgoan Industrial	High Heat + High Poverty
P3	Badda	High heat + Low poverty
P4	Cantonment	Low heat + High poverty
P5	Tejgoan Industrial	Special Case: Industrial Hotspot
P6	Derma	Low Heat + Low Poverty

Among these, Gulshan (P1) is used as the primary reference point because it exhibits a strong and consistent warming trend, substantial LULC transformation, and a unique combination of high heat exposure and high poverty concentration within informal settlements embedded in an otherwise affluent zone. This duality makes Gulshan an analytically rich case for understanding how LULC change and socioeconomic vulnerability interact to shape thermal environments.

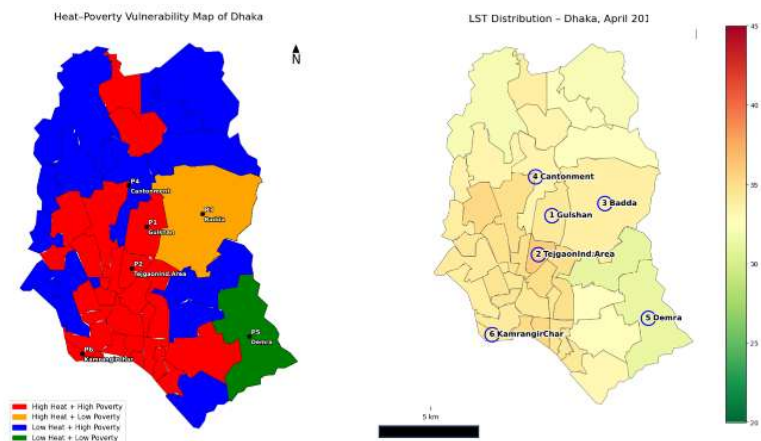


Figure 1c: Heat–Poverty Vulnerability Map of Dhaka (left) and LST Distribution for April 2017 (right).

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Detailed Case Study: Gulshan (P1)

High Heat + High Poverty

Multi-temporal satellite imagery (2014–2024) reveals substantial land-cover transformation in Gulshan, particularly around the lakeside informal settlements and rapidly densifying commercial corridors. In 2014, the area exhibited a mosaic of medium-density built-up structures, scattered vegetation, and open spaces. These conditions typically moderate surface temperature through shading and evapotranspiration (Oke, 1982; Voogt & Oke, 2003), which is reflected in the relatively lower LST values during the early years.

By 2017 and 2018, the imagery shows accelerated construction activity, expansion of paved surfaces, and reduction of vegetated patches. This corresponds with a noticeable rise in maximum LST, consistent with the well-established relationship between impervious surface expansion and surface heating (Weng, 2009; Zhang et al., 2014). The trend intensifies after 2020, when high-rise commercial buildings, glass façades, and metal roofing become dominant features. These materials have high heat storage capacity and low emissivity, contributing to elevated LST values (Li et al., 2013).

The time-series graph for Gulshan (Figure 2A) shows a clear upward trajectory in pre-monsoon maximum LST, with a statistically significant trend of $+0.51\text{ }^{\circ}\text{C}/\text{year}$. This is one of the steepest warming rates among the selected points, indicating rapid thermal intensification. The highest LST values (2022–2024) coincide with the period of maximum construction density and near-complete loss of vegetated pockets.

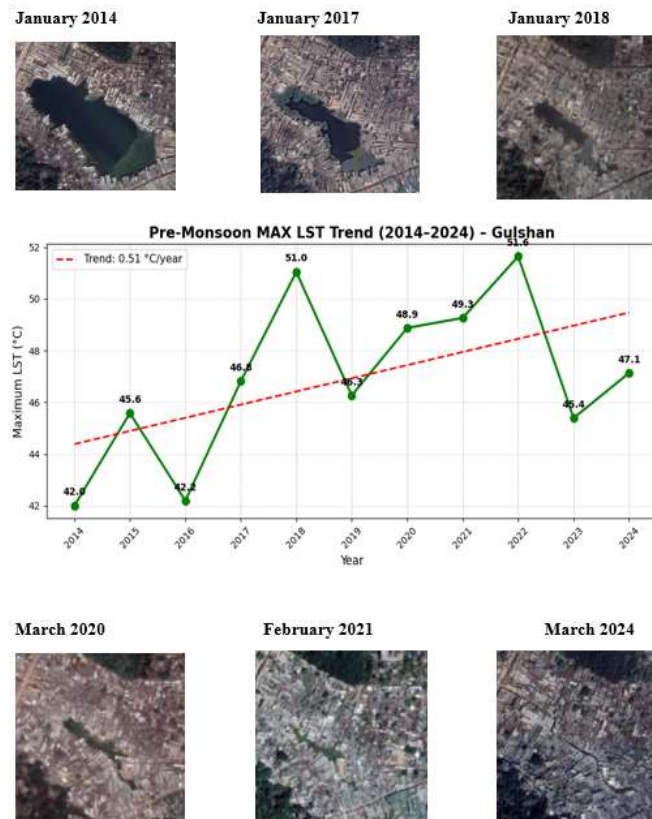


Figure 1d: LST and LULC Changes in Point 1 (Gulshan)

Source: GEE and Python processing, combined with Google Satellite time-lapse imagery. Map by Mahbubul Alam (2025).

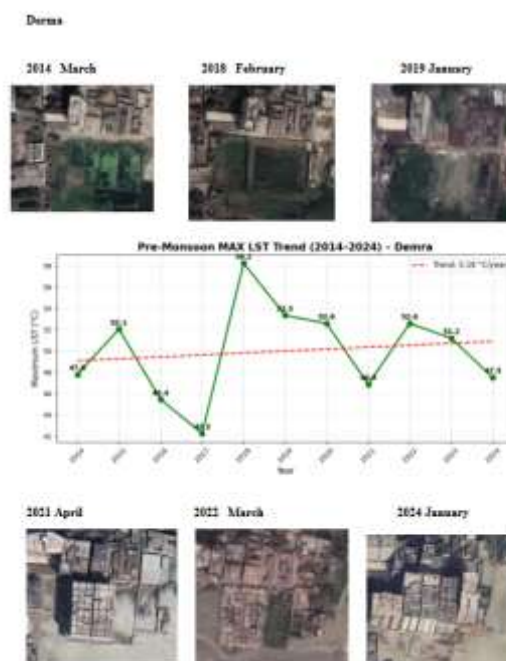


Figure 1e LST and LULC Changes in Point 5 (Demra)

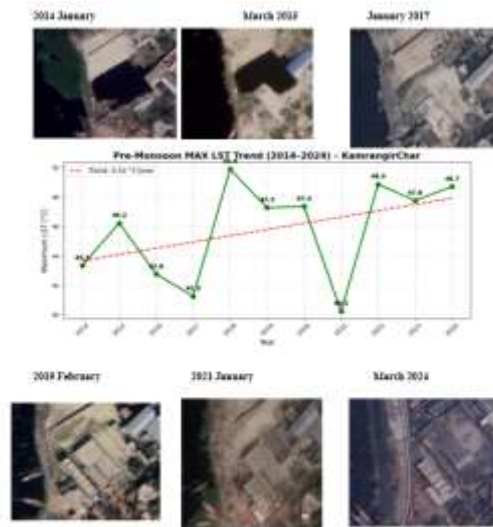


Figure 1f : LST and LULC Changes in Point 6 (Kamrangirchar)

This figure presents the Land Surface Temperature (LST) and land use/land cover (LULC) changes at Point 1 (Gulshan) Figure 2a. The blue line is the line graph of LST data in the time series, and the green dashed line is a statistical trendline showing the upward trend of LST at this specific point. The satellite images illustrate progressive urban expansion and reduction of vegetated surfaces between 2014 and 2024, contributing to the observed warming trend. Gulshan, categorized as High Heat + High Poverty, demonstrates significant thermal escalation linked to dense built-up growth and declining surface permeability

Importantly, Gulshan contains several high-poverty informal settlements located adjacent to high-value commercial and residential developments. These communities experience disproportionately high heat exposure due to poor housing materials, limited ventilation, and minimal tree cover—conditions known to amplify heat vulnerability (Harlan et al., 2006; IPCC, 2022). Thus, Gulshan exemplifies how urban inequality and land-cover change converge to produce localized heat stress. Heat–Poverty Vulnerability Map of Dhaka This map illustrates the spatial distribution of heat–poverty vulnerability across Dhaka, categorized into four classes. Zones P1–P6 are marked for detailed analysis. The map supports the interpretation of thermal and socioeconomic patterns. Here only Gulshan, Derma and Kamrangirchar.

1C Seasonal Heat-Risk Clustering

A detailed seasonal heat-risk clustering analysis using K-means ($k = 4$) was conducted for the pre-monsoon, monsoon, and winter seasons from 2014 to 2024. The results confirm that pre-monsoon months exhibit the highest and most spatially extensive heat-risk levels, monsoon months show moderated but persistent hotspots, and winter months display the lowest overall risk but retain structural hotspots in dense urban cores

To classify seasonal heat-risk zones, a composite dryness index (CDI) was calculated by subtracting normalized NDVI from normalized LST for each zone. This index captures the imbalance between surface heating and ecological buffering, with higher CDI values indicating hotter, less vegetated areas. Prior to clustering, both NDVI and LST values were min–max

normalized to ensure comparability across years and variables. K-means clustering ($k = 4$) was applied to the CDI–NDVI–LST feature space to group zones into four heat-risk categories: High Risk, Medium Risk, Lower-Medium Risk, and Low Risk. An additional “No Data” class was assigned to zones lacking valid satellite observations due to cloud cover, water surfaces, or insufficient pixel coverage.

Pre-Monsoon (March–May): Highest Heat-Risk Season

The Pre-Monsoon clustering maps exhibit the strongest and most spatially extensive heat-risk levels. High-risk clusters expand markedly across dense urban and industrial zones, including Tejgaon, Rampura, Mirpur, and Jatrabari. These areas consistently fall into the highest-risk category due to impervious surfaces, limited vegetation cover, and intense anthropogenic heat emissions.

Cooler clusters remain concentrated around Hatirjheel, Ramna Park, the Botanical Garden, and river corridors; however, these zones gradually shrink over the decade, indicating vegetation loss and increasing urban encroachment. Overall, the Pre-Monsoon season represents the most severe and widespread heat-stress period, making it the critical window for heat-risk management and early-season intervention.

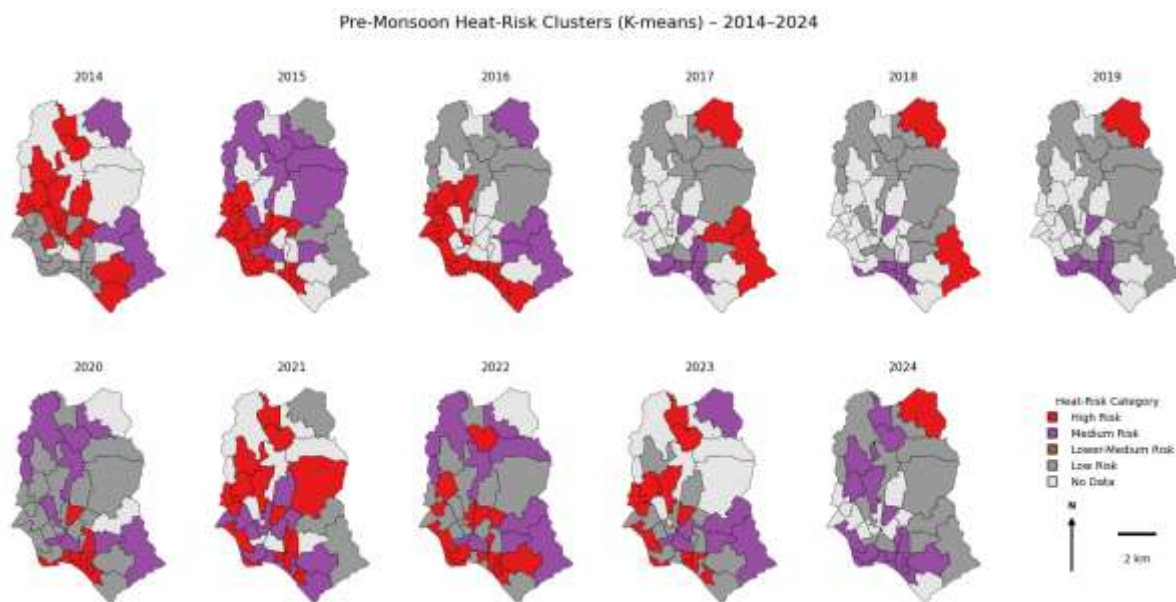


Figure 1g :Pre-Monsoon Heat-Risk Patterns

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Monsoon (June–September): Moderated but Persistent Urban Heat

Despite increased rainfall, cloud cover, and vegetation regrowth, Monsoon clustering results reveal persistent high-risk zones within the same dense urban cores—particularly Tejgaon, Rampura, Mirpur, and Jatrabari. Although overall temperatures decrease, the spatial pattern of heat risk remains structurally similar to the Pre-Monsoon season. Low-risk clusters become more fragmented, especially in peri-urban areas, suggesting that Monsoon cooling is

insufficient to counteract the thermal inertia of built-up surfaces. This season underscores the resilience of urban heat islands even under favourable climatic conditions, highlighting the dominance of land-cover structure over short-term climatic moderation.

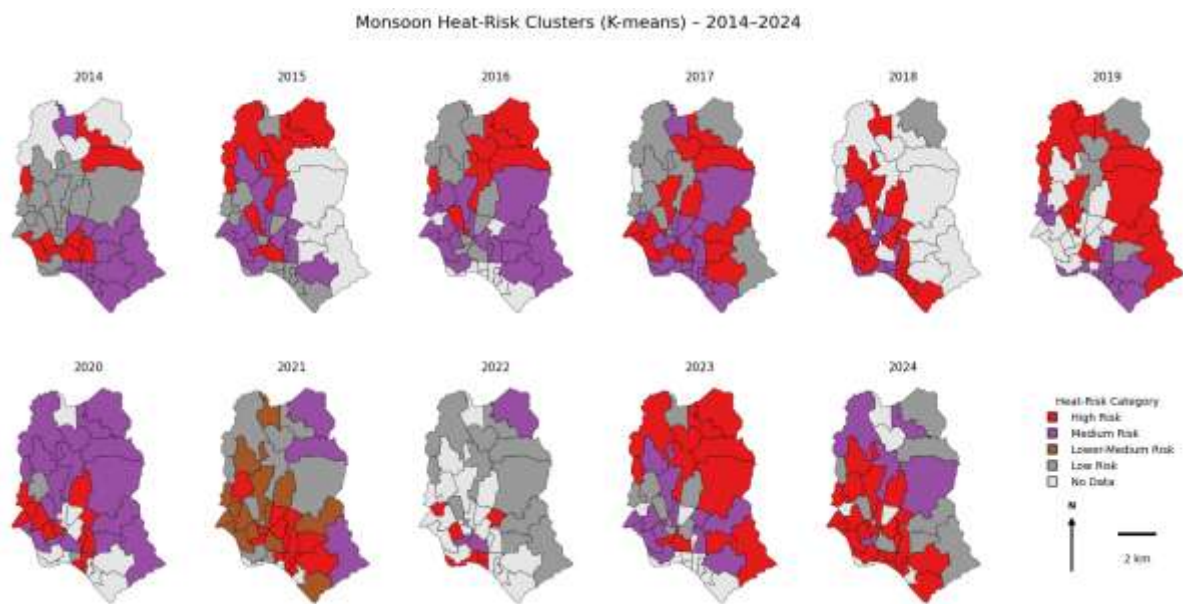


Figure 1g: Monsoon Heat-Risk Patterns

Source: GEE and Python processing. Map by Mahbubul Alam (2025).

Winter (October–February): Lowest Heat Risk but Persistent Hotspots

Winter clustering displays the lowest overall heat-risk levels, reflecting reduced solar radiation and cooler ambient temperatures. However, several dense urban cores—such as Tejgaon, Kamrangirchar, Mohammadpur, and parts of Mirpur—remain classified as high-risk zones.

Cool clusters are concentrated around major parks and waterbodies, but their spatial extent declines over the decade, indicating the gradual erosion of natural cooling buffers. Winter results demonstrate that structural heat hotspots persist year-round, even when climatic conditions are favourable, reinforcing the influence of built-up density and vegetation scarcity on thermal behaviour.

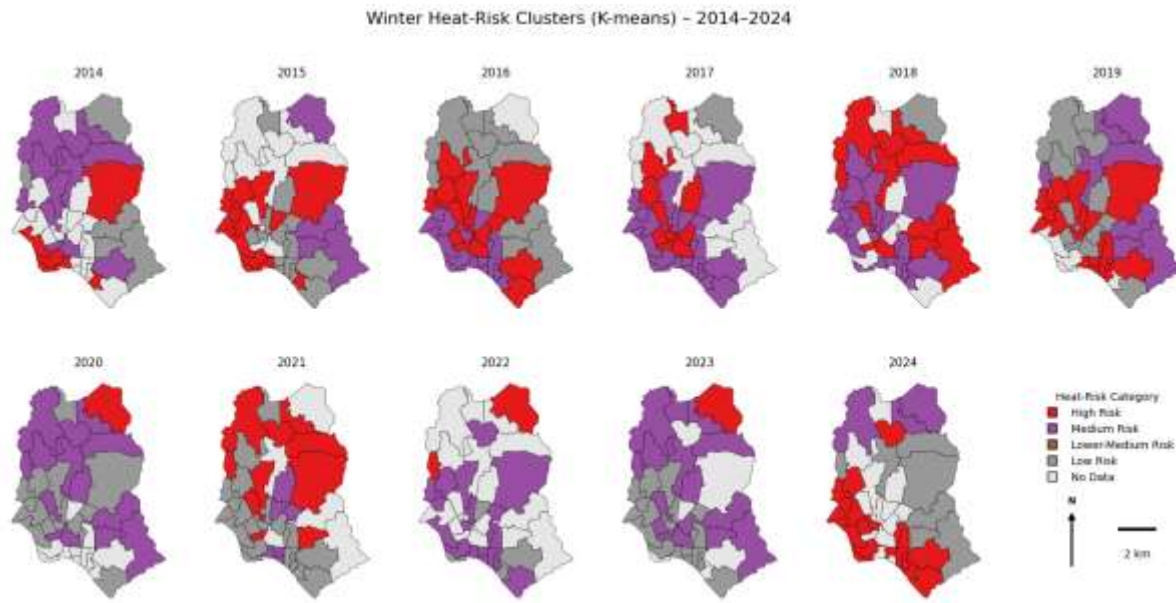


Figure 1h : Winter Heat-Risk Patterns

Across all seasons, Dhaka's heat-risk landscape is shaped by the combined effects of urban density, vegetation scarcity, and ongoing land-cover transformation.

1. Pre-Monsoon shows the highest intensity and widest spatial extent of heat risk.
2. Monsoon exhibits moderated temperatures but persistent structural hotspots.
3. Winter presents overall cooling but enduring high-risk zones in dense urban cores.

The progressive fragmentation of low-risk clusters and the shrinking of cool zones highlight the urgent need for season-specific heat-mitigation strategies. These include expanding urban greening, protecting cooling corridors, increasing reflective and permeable surfaces, and prioritizing interventions in structurally persistent hotspots.

*This map series presents seasonal heat-risk clusters derived from K-means classification of CDI, NDVI, and LST data from 2014 to 2024. Zones with insufficient satellite observations were assigned to a 'No Data' category to preserve methodological integrity. The layout includes a unified legend, directional arrow, and scale bar to support spatial interpretation and reproducibility."

1D. CDI-Based Spatial Contrast (2014–2024)

The Cooling Deficit Index (CDI), defined as normalized LST minus normalized NDVI, provides an integrated measure of thermal stress. CDI maps (34 show:

- High CDI (dark red) aligns with hotspot zones
- Low CDI (light tones) aligns with cold spot zones
- CDI intensifies over time, especially in central and industrial zones

This confirms that hotspots are not random but structurally embedded in Dhaka's urban form.

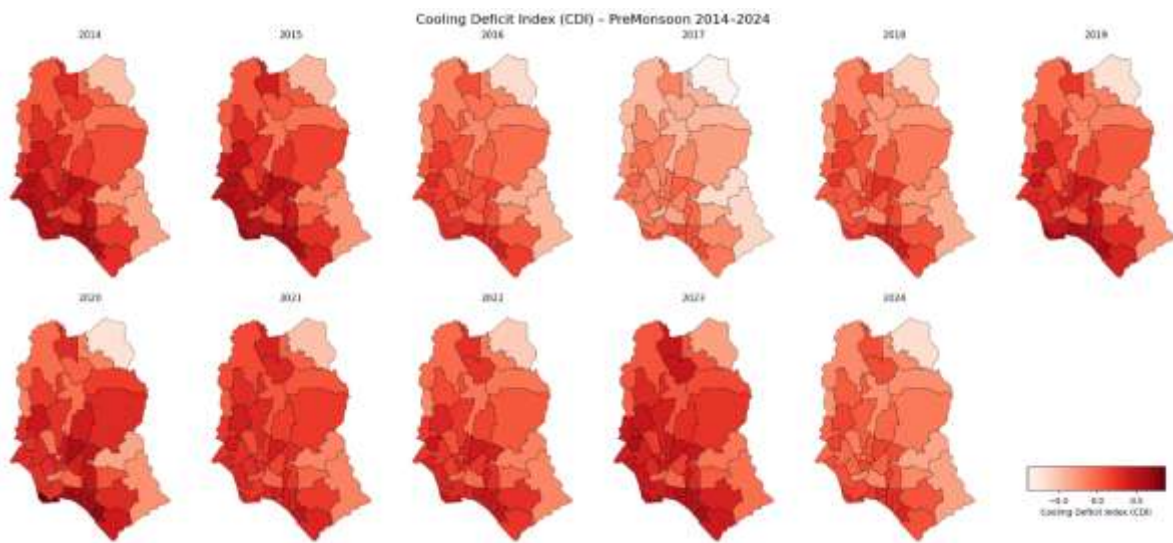


Figure 1i: Cooling Deficit Index (CDI) Maps for Dhaka (2014–2024)

Appendix 2

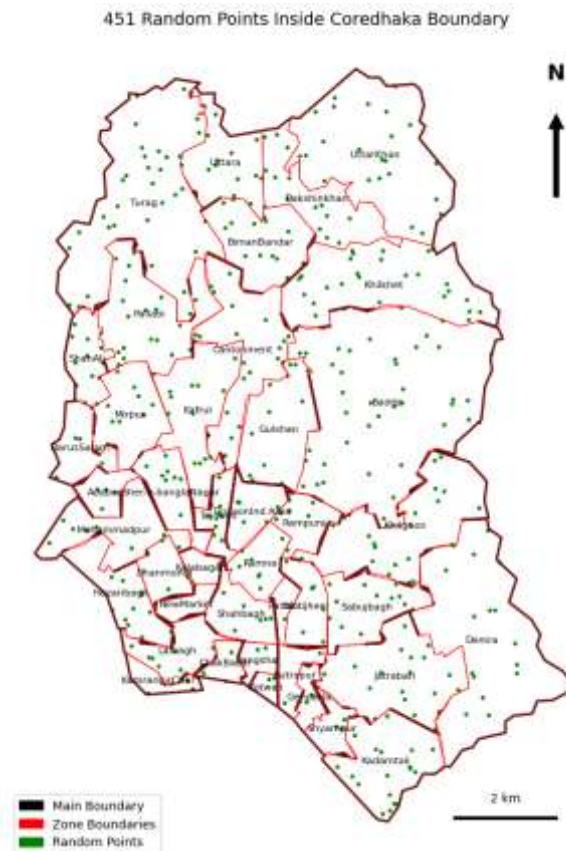


Figure : 2a

Appendix 3.

Table 1 : Zone- LST Difference (April – December, 2017)

	Zone	April_LST	December_LST	Delta_LST
22	Mirpur	35.254762	26.402143	8.852619
0	Adabor	34.385000	25.848750	8.536250
2	Bangshal	35.250000	26.746667	8.503333
37	TejgaonInd.Area	35.638889	27.221667	8.417222
5	ChakBazar	35.103333	26.700000	8.403333
9	Dhanmondi	34.646667	26.375000	8.271667
28	Ramna	34.883333	26.683333	8.200000

	Zone	April_LST	December_LST	Delta_LST
24	Motijheel	34.930000	26.753333	8.176667
36	Tejgaon	34.860000	26.687500	8.172500
33	Sher-e-banglaNagar	34.540000	26.380000	8.160000
23	Mohammadpur	33.917778	25.775833	8.141944
15	Kafrul	34.828333	26.753125	8.075208
11	Gulshan	34.624545	26.569091	8.055455
32	Shahbagh	34.254444	26.214167	8.040278
16	Kalabagan	34.330000	26.360000	7.970000
7	DarusSalam	33.642000	25.703000	7.939000
25	NewMarket	34.170000	26.235000	7.935000
27	Paltan	34.236667	26.355000	7.881667
29	Rampura	34.765556	26.983333	7.782222
31	ShahAli	33.003333	25.283333	7.720000
21	Lalbagh	34.118000	26.451000	7.667000
35	Sutrapur	34.750000	27.085000	7.665000
40	Uttara	33.782381	26.152143	7.630238
12	Hazaribagh	33.505000	25.882500	7.622500
17	KamrangirChar	33.930000	26.370000	7.560000
26	Pallabi	32.824359	25.432692	7.391667
4	Cantonment	33.237692	25.875769	7.361923
14	Kadamtali	33.324815	25.964444	7.360370
1	Badda	33.814912	26.534737	7.280175
3	BimanBandar	33.220370	26.106667	7.113704
10	Gendaria	34.296667	27.222500	7.074167
6	Dakshinkhan	33.154167	26.175625	6.978542

	Zone	April_LST	December_LST	Delta_LST
34	Shyampur	34.086667	27.130000	6.956667
20	Kotwali	33.736667	27.000000	6.736667
38	Turag	31.906543	25.244444	6.662099
30	Sabujbagh	32.669167	26.225625	6.443542
39	UttarKhan	31.968182	25.561364	6.406818
18	Khilgaon	32.354000	25.957667	6.396333
19	Khilkhet	32.159474	25.840789	6.318684
13	Jatrabari	32.931778	26.660000	6.271778
8	Demra	31.192609	25.388696	5.803913

Table 2 Theil–Sen Slope and MK trend

	Zone	MK Z	MK p	MK Trend	TS Slope per year
0	Adabor	-1.401298	0.161125	no trend	-0.091077
1	Badda	-0.778499	0.436275	no trend	-0.059925
2	Bangshal	-1.712698	0.086768	no trend	-0.117930
3	BimanBandar	0.155700	0.876270	no trend	0.048960
4	Cantonment	0.000000	1.000000	no trend	-0.000781
5	ChakBazar	-2.335497	0.019517	decreasing	-0.164237
6	Dakshinkhan	-0.467099	0.640429	no trend	-0.012561
7	DarusSalam	-0.311400	0.755497	no trend	-0.044796
8	Demra	0.311400	0.755497	no trend	0.036648
9	Dhanmondi	-2.024097	0.042960	decreasing	-0.157119
10	Gendaria	-2.024097	0.042960	decreasing	-0.090150
11	Gulshan	-1.089899	0.275758	no trend	-0.034891
12	Hazaribagh	-1.868397	0.061707	no trend	-0.156668
13	Jatrabari	-0.467099	0.640429	no trend	-0.041359
14	Kadamtali	-0.934199	0.350201	no trend	-0.033635
15	Kafrul	-0.934199	0.350201	no trend	-0.041958
16	Kalabagan	-1.556998	0.119471	no trend	-0.100611
17	KamrangirChar	-2.335497	0.019517	decreasing	-0.148690
18	Khilgaon	-0.311400	0.755497	no trend	-0.031920
19	Khilkhet	0.000000	1.000000	no trend	-0.028684
20	Kotwali	-2.024097	0.042960	decreasing	-0.126276
21	Lalbagh	-1.868397	0.061707	no trend	-0.133087
22	Mirpur	-1.401298	0.161125	no trend	-0.085372
23	Mohammadpur	-2.024097	0.042960	decreasing	-0.162148
24	Motijheel	-0.934199	0.350201	no trend	-0.075685
25	NewMarket	-1.089899	0.275758	no trend	-0.097360

26	Pallabi	-0.934199	0.350201	no trend	-0.047256
27	Paltan	-0.778499	0.436275	no trend	-0.055916
28	Ramna	-1.245598	0.212912	no trend	-0.094120
29	Rampura	-1.245598	0.212912	no trend	-0.085493
30	Sabujbagh	-0.467099	0.640429	no trend	-0.028976
31	ShahAli	-1.089899	0.275758	no trend	-0.046873
32	Shahbagh	-0.622799	0.533417	no trend	-0.055369
33	Sher-e-banglaNagar	-1.868397	0.061707	no trend	-0.108326
34	Shyampur	-2.179797	0.029273	decreasing	-0.068868
35	Sutrapur	-1.712698	0.086768	no trend	-0.110014
36	Tejgaon	-1.401298	0.161125	no trend	-0.082013
37	TejgaonInd.Area	-1.868397	0.061707	no trend	-0.098923
38	Turag	-0.622799	0.533417	no trend	-0.048017
39	UttarKhan	-0.467099	0.640429	no trend	-0.051373
40	Uttara	0.311400	0.755497	no trend	0.036531

TS_Lower_CI TS_Upper_CI

3 -0.203421 0.205550
8 -0.092412 0.111818
40 -0.134484 0.108440
4 -0.194972 0.132822
6 -0.209724 0.101264
19 -0.164067 0.113528
30 -0.184762 0.075504
18 -0.209238 0.099444
14 -0.222802 0.046670
11 -0.211942 0.074763
13 -0.150025 0.081341
15 -0.244440 0.071247
7 -0.191127 0.058066
31 -0.148511 0.072236
26 -0.219557 0.098763
38 -0.177105 0.111520
39 -0.218313 0.087471
32 -0.297118 0.062644
27 -0.301428 0.045175

1	-0.208751	0.087521
34	-0.211885	-0.004489
24	-0.299488	0.061017
36	-0.252631	0.044592
22	-0.297605	0.053701
29	-0.256192	0.060027
10	-0.310994	-0.008119
0	-0.235396	0.034153
28	-0.326940	0.059150
25	-0.245821	0.075254
37	-0.262829	0.000072
16	-0.257324	0.039142
33	-0.277177	0.014661
35	-0.342151	0.001314
2	-0.274782	0.018958
20	-0.286389	-0.005309
21	-0.260303	0.014719
17	-0.221960	-0.031320
12	-0.280572	0.015495
9	-0.285255	-0.007896
23	-0.299465	-0.019670
5	-0.299359	-0.019839

